

Language Equivalence Between Deterministic and Nondeterministic One-Counter Nets Is Decidable

Alp Eren Bütün

Independent Researcher, Türkiye

ORCID: 0009-0001-2479-6037

github.com/alperenbutun

September 2026

Abstract

We prove decidability of language equivalence between a deterministic one-counter net and a nondeterministic one-counter net under finite-word acceptance by final control state, with initial counter 0. The deterministic-right inclusion is a known decidable case; the difficult direction is treated under the reverse-inclusion promise. The proof combines exact forward envelopes and backward completion requirements with a shortest-counterexample analysis. Bounded deterministic height is controlled by a Dickson well-quasi-order. Unbounded behavior exposes bounded synchronized event cores, whose source-sink geometry is constrained by the reverse-inclusion promise. A full-envelope differential analysis then controls arbitrary mode switching inside bounded-return high excursions by a single combined useful-mode potential. Contextual power and arithmetic recurrences are converted into bounded local continuations, endpoint-preserving deletions, or promise contradictions; no global bound on an inherited suffix is used. Terminal unmatched growth is handled by a last-occurrence level ladder, and the remaining global behavior is organized as an effectively finitely branching tree of exact low-band states. Dickson’s lemma and König’s lemma make this continuation tree finite and yield a total computable small-witness bound for the hard inclusion. Combining it with deterministic-right inclusion gives a decision procedure for language equivalence. No complexity upper bound for the resulting equivalence procedure is claimed.

Keywords. one-counter nets; language equivalence; language inclusion; counter automata; residual semantics; max-plus systems; well-quasi-orders; decidability.

1 Introduction

A one-counter net (OCN) is a finite-state machine equipped with one nonnegative counter and no explicit zero test. The proof is developed in the standard unit-step normal form, where transitions change the counter by $-1, 0$, or $+1$. Section 2 gives an explicit language-preserving normalization from arbitrary integer transition weights to this form, preserving determinism and nondeterminism. Thus the final theorem covers the integer-weight formulation of Problem 24.4(B), not only its unit-step presentation.

Hofman and Totzke proved that trace inclusion of an OCN into a deterministic OCN is decidable, in fact NL-complete in their unit-weight model, while unrestricted trace inclusion for OCNs is undecidable [1]. Deterministic one-counter automata also admit efficient equivalence checking [2]. The mixed deterministic/nondeterministic *language equivalence* problem for one-counter nets, with acceptance by final state, has nevertheless remained open in the formulation recorded as Problem 24.4(B) in the Automata Exchange collection [4].

The main theorem of this paper is the following.

Theorem 1.1 (Main theorem). *Given a deterministic one-counter net D and a nondeterministic one-counter net N over the same finite alphabet, with arbitrary integer transition weights, both started with counter value 0 and accepting by final control state, it is decidable whether*

$$L(D) = L(N).$$

The proof separates the two inclusions. The direction $L(N) \subseteq L(D)$ is a known deterministic-right inclusion case: the compact-witness method of Hofman–Totzke applies to monotone relations including languages defined by final-state acceptance. The difficult direction is $L(D) \subseteq L(N)$. We solve it under the promise $L(N) \subseteq L(D)$ and prove a computable small-witness theorem:

$$L(D) \not\subseteq L(N) \implies \exists w \in L(D) \setminus L(N), \quad |w| \leq B(D, N),$$

for a total computable function B .

The hard direction is organized around two exact residual objects. For a prefix u , the forward envelope $H_u(q)$ is the largest counter value with which the nondeterministic net can reach state q after reading u . For a suffix v , the backward requirement $R_v(q)$ is the least counter value from which v can lead from q to acceptance. They satisfy

$$uv \in L(N) \iff \exists q \ H_u(q) \geq R_v(q).$$

Thus a rejected word has, at every cut, a strict finite-dimensional separator

$$H_u < R_v.$$

The proof proceeds through five interacting layers.

First, if the deterministic counter remains bounded, a well-quasi-order argument gives an effective bound on shortest counterexamples. Hence only high-counter behavior can obstruct decidability.

Second, sufficiently long high-counter regions expose a finite universe of bounded synchronized event cores. Under the promise $L(N) \subseteq L(D)$, recurrent positive and negative counter effects satisfy source–sink efficiency inequalities. Rational potentials, cone separation, and critical max-plus graphs then localize hidden credit to bounded-entry episodes.

Third, selected-run arguments are lifted to the *full* nondeterministic envelope. We analyze the synchronized triple graph $Q_D \times Q_N \times Q_N$, obtain one-sign differential potentials, isolate mixed-sign oscillation, and reduce the remaining lag/catch-up behavior to one finite combined useful-mode graph. Absence of positive useful cycles gives a max-plus potential that bounds all suffix-relevant forward-envelope coordinates throughout an arbitrary interleaving.

Fourth, local recurrence outcomes are made context safe. Zero- D powers are bounded by exact-forward self-cover; negative- and zero-mean recurrences are treated as bounded local transients or deterministic funding; and strict/exact source–sink routes are bounded locally or admit endpoint-preserving balanced deletion. Thus no argument has the invalid form “bounded local seed plus an unbounded inherited suffix.”

Fifth, the global proof uses two well-quasi-order trees rather than a finite event-cycle cap. A terminal high tail is decomposed by last occurrences of deterministic counter levels. Between successive ladder levels there is one +1 step followed by a same-level bounded-return excursion. The nonterminal part is organized by exact LOW states (p, d, H) , ordered by componentwise comparison of the full forward envelope. Every shortest counterexample induces a bad sequence, while node-dependent local bounds make the successor relation effectively finite. Dickson plus König therefore yields a finite weighted continuation tree and a computable hard-direction witness bound.

We deliberately do not claim a useful complexity upper bound. The computed witness bound is effective but may be enormous.

1.1 Research provenance

This paper is the next step in a continuous research program rather than an isolated attack on one-counter equivalence. The program began with a completion-aware 2:1 deterministic pushdown construction and its $p:q$ generalization, then moved through completion/quotient geometry and semantic/LIFO lift theory toward representation-independent continuation invariants [7, 6, 8, 11]. The later Prime-DOCA and Two-Factor DOCA papers supplied complementary canonical-cut, quotient-section, and last-safe-cut mechanisms in the one-counter setting [9, 10]. The present work replaces the earlier scalar residual by the exact forward envelope H and backward requirement R and develops the new full-envelope, bounded-return, terminal-ladder, and LOW-tree machinery required for nondeterministic OCN equivalence. This research lineage is methodological: apart from the external deterministic-right inclusion theorem identified in Section 18, the proof below is self-contained and does not use the author’s earlier theorems as black boxes.

1.2 Proof architecture

The proof is not a determinization argument. It is a shortest-counterexample compactness argument that progressively removes every possible source of unbounded credit. Table 1 summarizes the main layers.

Layer	Main object	Role in the proof
Residual semantics	Forward envelope H and backward requirement R	Gives the exact separator $H < R$ at every rejected cut and exact block transformers.
Bounded geometry	Deterministic height, boundary signatures, synchronized simple cycles	Compresses bounded-height runs and exposes a finite universe of primitive event cores.
Promise geometry	Positive sources, negative sinks, rational slope potentials	Uses $L(N) \subseteq L(D)$ to forbid over-efficient source/sink combinations and to localize hidden funding.
Full-envelope differential geometry	Triple-product strand differences and combined useful-mode graph	Controls arbitrary mode switching and bounds suffix-relevant forward credit in same-level bounded-return excursions.
Context-safe recurrence	Exact-forward self-cover, critical relations, causal source–sink matching	Replaces fixed-suffix power/arithmetic outcomes by bounded local continuation, shortening, or promise contradiction.
Global compactness	Terminal last-occurrence ladder and exact LOW-state WQO tree	Produces effective finite branching, finite bad-sequence trees, and the final computable small-witness bound.

Table 1: Logical architecture of the hard inclusion proof.

The key point is that each layer is needed because the preceding one leaves a genuine obstruction. Local recurrence alone is insufficient when a suffix can be funded by arbitrarily large incoming credit; projective recurrence alone is insufficient when absolute credit can escape; and a structural pump alone is insufficient for decidability unless its first usable seed is also effectively bounded. The final proof therefore combines local deletion, source–sink balancing, SCC resource rankings, and a finite-state recurrence argument.

1.3 Contributions

The main contributions are the following.

- (i) Exact forward-envelope and backward-requirement semantics for OCN final-state languages, including fixed-factor residuation.

- (ii) Effective bounded-height compression and protected-credit exposure for shortest counterexamples.
- (iii) A finite bounded event-core universe together with connector, boundary, and anchored-relative reductions.
- (iv) Promise-induced source-sign and source-sink efficiency theorems, converted into rational SCC budgets and exact critical-slope relations.
- (v) A full-envelope differential theory: one-sign strand potentials, bounded owner switching, canonical positive-relative source cuts, and a lag/catch-up sign reduction.
- (vi) A combined useful-mode potential that is invariant under arbitrary interleaving of the remaining neutral/losing event types inside a bounded-return excursion.
- (vii) Context elimination for every recurrent local family used in the proof: positive returns, neutral bounded expressions, negative mean, zero mean, and strict/exact source-sink routes.
- (viii) A terminal last-occurrence ladder theorem and a node-dependent exact LOW-state WQO construction, yielding a total computable hard-direction small-witness bound without a uniform global high-excursion bound.
- (ix) Composition with deterministic-right final-state inclusion to decide language equivalence.

1.4 Organization

Section 2 fixes the OCN model and acceptance convention, and Section 3 develops exact residual semantics. Sections 4–8 establish bounded-height compression, event extraction, full-phase recurrence, boundary control, and anchored relative geometry. Sections 9–13 derive promise-induced source-sink geometry, SCC budgets, drain classification, protected-source localization, and bounded-entry deferral. Sections 14–15 give the full-envelope bounded-return and context-elimination theorems. Section 16 treats terminal unmatched growth by a last-occurrence ladder. Section 17 constructs the exact LOW continuation tree and proves the hard-direction small-witness theorem. Sections 18–19 verify the deterministic-right inclusion input and derive equality decidability. The appendices collect the effectivity ledger and two bookkeeping points: unbounded absolute backward requirements and overlapping event occurrences.

2 One-counter nets and acceptance

Definition 2.1 (One-counter net). An OCN is a tuple

$$A = (Q, \Sigma, \delta, q_0, F),$$

where Q is a finite set of control states, Σ a finite alphabet, $q_0 \in Q$ an initial state, $F \subseteq Q$ a set of final states, and

$$\delta \subseteq Q \times \Sigma \times \{-1, 0, +1\} \times Q.$$

A configuration is a pair $(q, c) \in Q \times \mathbb{N}$. A transition

$$(q, c) \xrightarrow{a} (r, c + d)$$

is enabled if $(q, a, d, r) \in \delta$ and $c + d \geq 0$.

A run on $w = a_1 \cdots a_k$ is the obvious sequence of enabled transitions. We write

$$(q, c) \xrightarrow{w} (r, c')$$

when such a run exists.

Definition 2.2 (Final-state language). The language of A is

$$L(A) = \{w \in \Sigma^* : (q_0, 0) \xrightarrow{w} (f, c) \text{ for some } f \in F, c \in \mathbb{N}\}.$$

An OCN is *deterministic* if from every control state and input letter there is at most one outgoing transition. We use D for the deterministic net and N for the nondeterministic net.

For comparison with the integer-weight formulation of Problem 24.4(B), call an OCN *integer weighted* when the transition component d is allowed to range over $\mathbb{Z}$ rather than $\{-1, 0, +1\}$.

Lemma 2.3 (Unit-step normalization). *Every finite integer-weight OCN A can be effectively transformed into a unit-step OCN A^u over the same input alphabet such that*

$$L(A^u) = L(A).$$

The transformation preserves determinism and nondeterminism. Hence language equivalence for arbitrary integer-weight OCNs reduces effectively to language equivalence for unit-step OCNs.

Proof. Let W be the maximum absolute transition weight of A and choose an integer $M > W$ (take $M = 1$ when all weights are 0). Represent an original counter value uniquely as

$$n = Mc + r, \quad c \in \mathbb{N}, \quad 0 \leq r < M.$$

The unit-step machine stores the remainder r in finite control and the quotient c in its physical counter. Thus its states are pairs (q, r) .

For an original transition (q, a, z, q') and a remainder r , write

$$r + z = kM + r', \quad 0 \leq r' < M,$$

where $k = \lfloor (r + z)/M \rfloor$. Since $|z| < M$ and $0 \leq r < M$, one has $k \in \{-1, 0, +1\}$. Add the unit-step transition

$$((q, r), a, k, (q', r')).$$

If $k = -1$ and the quotient counter is 0, this transition is disabled by ordinary OCN underflow; this is exactly the case $n + z < 0$. In every other case the encoded step is enabled exactly when the original step is enabled, and the invariant $n = Mc + r$ is preserved.

The initial encoded configuration is $((q_0, 0), 0)$, and every (q, r) with $q \in F$ is final. Induction on the input length gives a one-to-one correspondence between feasible original runs and feasible encoded runs over the same word. Determinism is preserved because for each original state and letter there is at most one transition, and the new remainder component only refines the source state. The construction is finite and effective. It may enlarge the finite control by a factor M , which is irrelevant here because no complexity upper bound is claimed. $\square$

From this point through Section 17 we work in the unit-step normal form. Lemma 2.3 is applied once at the beginning for arbitrary integer-weight inputs. The use of unit transitions is also standard in the OCN literature; succinct integer-weight presentations have the same expressive power [3].

Lemma 2.4 (Monotonicity). *If*

$$(q, c) \xrightarrow{w} (r, d)$$

and $c' \geq c$, then

$$(q, c') \xrightarrow{w} (r, d + c' - c).$$

Consequently, for fixed q the set of counter values from which a fixed word is executable is upward closed.

Proof. Reuse the same transition sequence. Every intermediate counter value is increased by the constant $c' - c$. $\square$

3 Exact residual semantics

For a word u and state $q \in Q_N$, define

$$H_u(q) = \max\{h \in \mathbb{N} : (q_0, 0) \xrightarrow{u} (q, h)\},$$

with $H_u(q) = \perp$ if q is unreachable. We order $\perp < h$ for all $h \in \mathbb{N}$.

For a word v and state q , define

$$R_v(q) = \min\{h \in \mathbb{N} : (q, h) \xrightarrow{v} (f, c) \text{ for some } f \in F, c \in \mathbb{N}\},$$

with $R_v(q) = \infty$ if no such h exists.

Proposition 3.1 (Exact cut semantics). *For every factorization $w = uv$,*

$$uv \in L(N) \iff \exists q \in Q_N : H_u(q) \geq R_v(q).$$

Hence

$$uv \notin L(N) \iff H_u(q) < R_v(q) \text{ for every } q,$$

using the conventions $\perp < r$ and $h < \infty$.

Proof. If uv is accepted, cut an accepting run after u ; its state q and counter h satisfy $h \leq H_u(q)$ and $h \geq R_v(q)$. Conversely, if $H_u(q) \geq R_v(q)$, take a prefix run realizing $H_u(q)$ and use monotonicity to execute an accepting v -run that needs only $R_v(q)$. $\square$

For a fixed word x , define the forward transformer

$$(T_x H)(r) = \max_{\substack{q, \pi: q \xrightarrow{x} r \\ H(q) \geq \text{req}(\pi)}} (H(q) + \text{eff}(\pi)),$$

where $\text{eff}(\pi)$ is the total counter effect of π and $\text{req}(\pi)$ is the least initial counter needed to execute π . Similarly define the backward transformer

$$(S_x R)(q) = \min_{r, \pi: q \xrightarrow{x} r} \max(\text{req}(\pi), R(r) - \text{eff}(\pi)).$$

Theorem 3.2 (Exact residuation). *For every forward envelope H , backward requirement R , and word x ,*

$$T_x(H) < R \iff H < S_x(R),$$

componentwise.

Proof. Fix q . The inequality $H(q) < S_x(R)(q)$ means that for every x -path $\pi : q \rightarrow r$, either $H(q) < \text{req}(\pi)$ or $H(q) + \text{eff}(\pi) < R(r)$. This is exactly the statement that no contribution from q can make the r -coordinate of $T_x(H)$ reach $R(r)$. Taking all q, r gives the equivalence. $\square$

Remark 3.3. The pair (H, R) is the basic residual object of the proof. It is not a scalar balance: it is a finite-dimensional available-credit/completion-demand separator.

4 Shortest counterexamples and bounded height

Assume for the remainder of the hard-direction proof that

$$L(N) \subseteq L(D)$$

and let $w \in L(D) \setminus L(N)$ be shortest.

Definition 4.1 (Local recurrence outcome). A *local recurrence outcome* is finite structural data extracted from a bounded macro configuration which yields one of the following: an endpoint-preserving deletion, a contradiction to the standing promise, a bounded local continuation/transfer, or a repeatable power/arithmetic family relative to the current continuation support. For brevity, an interval is called *certificate-free* when none of these local recurrence outcomes has yet been extracted.

This terminology is purely local. A repeatable family that retains an inherited suffix is *not* by itself treated as a bounded global witness. Section 15 proves that every contextual power/arithmetic outcome used below is instead absorbed into bounded local continuation, shortening, or promise contradiction before the global witness bound is assembled.

Let the unique run of D on w visit deterministic configurations

$$(p_t, d_t), \quad t = 0, \dots, |w|.$$

Lemma 4.2 (Mixed-domination deletion). *Suppose two cuts $i < j$ have the same deterministic control state,*

$$p_i = p_j,$$

and

$$d_i \geq d_j, \quad H_i \leq H_j.$$

Then deleting the factor between i and j yields a shorter word still in $L(D) \setminus L(N)$.

Proof. The deleted deterministic suffix starts from at least the original height d_j , so the same deterministic suffix remains feasible. On the nondeterministic side, replacing H_j by the smaller envelope H_i cannot create acceptance of the fixed suffix, because the original cut satisfies $H_j < R_j$. $\square$

The next fact is the first compactness theorem.

Theorem 4.3 (Effective bounded-height compression). *For every fixed bound B , there is a total computable*

$$F(B, D, N)$$

such that if w is a shortest word in $L(D) \setminus L(N)$ and

$$d_t \leq B \quad \text{for all } t,$$

then

$$|w| \leq F(B, D, N).$$

Proof. Encode $\perp$ by 0 and a finite envelope value h by $h + 1$; write $\hat{H}_t \in \mathbb{N}^{Q_N}$ for the resulting vector. At cut t define

$$X_t = (p_t, B - d_t, \hat{H}_t).$$

For a fixed deterministic control state, order these records componentwise in the last $1 + |Q_N|$ coordinates. The disjoint union over the finitely many control states is a well-quasi-order by Dickson's lemma.

If $i < j$ and $X_i \leq X_j$, then $p_i = p_j$,

$$B - d_i \leq B - d_j \iff d_i \geq d_j,$$

and $H_i \leq H_j$. Lemma 4.2 would delete the factor between the two cuts. Hence the cut sequence of a shortest counterexample is a bad sequence for this well-quasi-order.

Effectivity is slightly stronger than mere well-quasi-ordering. At depth t , every finite envelope coordinate is at most t because the initial counter is 0 and every transition has effect at most $+1$.

Thus every coordinate of $\hat{H}_t$ is at most $t + 1$. Consider the rooted tree of all finite bad sequences satisfying these depth-dependent bounds and the finite physical-color restriction $d_t \in \{0, \dots, B\}$. The tree is finitely branching and effectively generated from D, N, B . Dickson's lemma forbids an infinite branch, so König's lemma implies that this effective tree is finite. Breadth-first generation eventually exhausts it and computes its maximum depth; call this maximum $F(B, D, N)$. $\square$

5 Protected exposure and bounded event cores

The hard case is therefore a shortest counterexample with large deterministic height.

Lemma 5.1 (Positive-credit exposure). *Let x be a factor between two cuts with the same deterministic control state. Suppose*

$$\delta_D(x) = c > 0, \quad H_{\text{before}} \leq H_{\text{after}}.$$

Then, in a shortest counterexample, the deterministic suffix after x must later visit a height at most $c - 1$.

Proof. If the suffix stayed at height at least c , deleting x would lower the entire suffix by c but keep it nonnegative. The deterministic word would still be accepted. The nondeterministic envelope before x is no larger than after x , so the deleted word would remain rejected, contradicting shortestness. $\square$

The proof repeatedly extracts short synchronized cycles. Put

$$M_2 = |Q_D||Q_N|, \quad M_3 = |Q_D||Q_N|^2.$$

A simple cycle in the synchronized pair graph has length at most M_2 , and a simple cycle in the synchronized triple graph has length at most M_3 .

Definition 5.2 (Primitive event cores). The primitive event universe consists of:

- (i) simple deterministic control cycles with nonzero effect;
- (ii) synchronized D/N pair cycles of length at most M_2 with nonzero relative effect

$$\Delta_{\text{rel}} = \Delta_N - \Delta_D;$$

- (iii) synchronized $D/N/N$ triple cycles of length at most M_3 with nonzero strand-differential effect;
- (iv) bounded boundary/support events.

All primitive core words have effectively bounded length; write B_{core} for a common bound.

5.1 Backward spread and finite connector phase

For a finite requirement vector R on support A , write

$$\text{spread}(R) = \max_{q, r \in A} |R(q) - R(r)|.$$

Lemma 5.3 (Backward differential escape). *Let*

$$C_R = 2(M_3 - 1).$$

Along a connector segment with stable finite backward support, if

$$\text{spread}(R) > C_R,$$

then two Bellman-optimal completion strands expose a synchronized $D/N/N$ triple cycle with nonzero N -strand differential effect.

Proof. Choose two finite coordinates realizing a gap larger than C_R and follow Bellman-optimal strands through the connector. Record the synchronized triple consisting of the deterministic control state and the two N control states. If every repeated triple had zero differential effect between the two N strands, erase the corresponding closed subwalks. The remaining triple path is simple, has length at most $M_3 - 1$, and each step changes the strand gap by at most 2. Hence the total gap change is at most $2(M_3 - 1)$, a contradiction. $\square$

Corollary 5.4 (Finite backward projective phase). *If a connector exposes neither a boundary event nor a triple differential core, then its normalized backward requirement profile ranges over an effectively finite set, independently of the absolute requirement level.*

Proof. The support is finite, and Lemma 5.3 bounds every pairwise finite-coordinate difference by C_R . Normalize by subtracting the minimum finite coordinate. $\square$

Theorem 5.5 (Connector compression). *There is a total computable constant B_{conn} such that any segment of a shortest counterexample that exposes none of*

- (a) *an N-boundary event,*
- (b) *a bounded differential core,*
- (c) *a bounded nonzero-D sign core,*
- (d) *an already available pump/deletion certificate,*

has length $< B_{\text{conn}}$.

Proof. By Corollary 5.4, throughout the connector the normalized finite part of the backward requirement belongs to an effectively finite set. Refine it to the connector color

$$\kappa_t = (p_t, A_t, R_t - \min_{A_t} R_t \mathbf{1}_{A_t}, \text{threshold/support regime}),$$

where $A_t = \{q : R_t(q) < \infty\}$. The color set is finite.

Suppose $\kappa_i = \kappa_j$ for $i < j$, and let W be the intervening word. The deterministic control state is the same at both ends. If $\Delta_D(W) \neq 0$, a closed deterministic walk of nonzero total effect contains a simple nonzero-effect deterministic cycle, contrary to the connector hypotheses. Hence $\Delta_D(W) = 0$.

Equal projective backward profiles give

$$R_j = R_i + \beta \mathbf{1}_A$$

on the common finite support A , so R_i and R_j are componentwise comparable. Exact backward semantics gives $R_i = S_W(R_j)$. If $R_i \leq R_j$, deleting W preserves deterministic feasibility and the strict separator, contradicting shortestness. If $R_j \leq R_i = S_W(R_j)$, then W is a zero-effect backward-demand self-cover; monotonicity of S_W yields a pump certificate. Thus no connector color can repeat, and the number of cuts is bounded by the finite color count. This count is an effective choice of B_{conn} . $\square$

Corollary 5.6 (Event-skeleton normal form). *Every shortest counterexample decomposes into bounded event words separated by connectors of length $< B_{\text{conn}}$, except when a local recurrence outcome has already been extracted.*

6 Interior full phases and sign locking

The event analysis repeatedly uses intervals on which support and threshold regimes are stable. We isolate the exact recurrence used later.

Definition 6.1 (Compatible full phase). Two interior cuts $i < j$ are in the same compatible full phase if:

- (i) the deterministic control states agree;
- (ii) the forward supports E_i, E_j agree;
- (iii) the finite backward supports A_i, A_j agree;
- (iv) the normalized forward profiles agree on E_i ;
- (v) the normalized backward profiles agree on A_i ;
- (vi) the finite threshold/regime labels agree.

Then there are integers α, β such that

$$H_j = H_i + \alpha \mathbf{1}_E, \quad R_j = R_i + \beta \mathbf{1}_A$$

on their finite supports.

Let W be the intervening factor and

$$e = \Delta_D(W).$$

Theorem 6.2 (Full-phase sign lock). *In a shortest certificate-free counterexample, a nontrivial compatible full-phase return satisfies*

$$\text{sgn}(e) = \text{sgn}(\alpha) = \text{sgn}(\beta) \neq 0.$$

Thus:

$$e > 0 \Rightarrow \alpha > 0, \beta > 0, \quad e < 0 \Rightarrow \alpha < 0, \beta < 0,$$

and $e = 0$ cannot survive.

Proof. We first prove the forward sign.

If $e > 0$ and $\alpha \leq 0$, then

$$T_W(H_i) = H_j \leq H_i.$$

One executable copy of W raises the deterministic height, hence all further copies are executable and the fixed suffix remains deterministically feasible. Monotonicity gives

$$T_W^k(H_i) \leq T_W(H_i) = H_j \quad (k \geq 1).$$

Since the original word is rejected at cut j ,

$$H_j < R_j,$$

so every word $uW^k v$ is in $L(D) \setminus L(N)$, contradicting certificate-freeness.

If $e < 0$ and $\alpha \geq 0$, deleting W starts the deterministic suffix from a higher counter value. Moreover

$$H_i \leq H_j < R_j,$$

so the deleted word remains N-rejected, contradicting shortestness.

If $e = 0$ and $\alpha \geq 0$, the same deletion argument applies. If $e = 0$ and $\alpha < 0$, then W is a neutral forward-contraction return; iterating it preserves deterministic feasibility and yields a neutral pump family. Hence no zero-effect same-phase return survives.

Now prove the backward sign. Exact backward semantics gives

$$R_i = S_W(R_j), \quad R_j = R_i + \beta \mathbf{1}_A.$$

If $e > 0$ and $\beta \leq 0$, then

$$R_j \leq R_i = S_W(R_j).$$

Thus W is a backward-demand self-cover at phase R_j . The deterministic side can repeat W indefinitely, and monotonicity of S_W yields

$$R_j \leq S_W(R_j) \leq S_W^2(R_j) \leq \dots$$

Since the original separator satisfies

$$H_i < S_W(R_j),$$

every further copy preserves rejection, giving an infinite $D \setminus N$ family.

If $e < 0$ and $\beta \geq 0$, then $R_i \leq R_j$. Deleting W starts the deterministic suffix higher, and

$$H_i < R_i \leq R_j$$

shows that the shortened word remains rejected by N .

The case $e = 0$ has already been excluded by the forward argument. Therefore

$$\text{sgn}(e) = \text{sgn}(\alpha) = \text{sgn}(\beta) \neq 0.$$

□

Remark 6.3. The theorem is used only on genuine interior full phases. Equality of an *active-only* backward projection is not sufficient and is never used as a substitute for this hypothesis.

Corollary 6.4 (Large clean excursion). *Inside a clean interior corridor with finitely many full-phase colors, a sufficiently large deterministic rise contains a positive full-phase return, and a sufficiently large deterministic fall contains a negative full-phase return.*

Proof. Take first-hit cuts of successive deterministic heights. More height levels than full-phase colors force equal colors at distinct heights. Theorem 6.2 determines the corresponding scalar signs. □

7 Boundary and low-active branches

Boundary recurrence is used only as a local reduction; we do not require a uniform D, N -only count at unbounded entry height.

Proposition 7.1 (Boundary recurrence reduction). *Refine a boundary occurrence by its deterministic control state, support data, and bounded forward boundary profile. For two consecutive occurrences of the same refined signature, exactly one of the following occurs:*

- (i) *the intervening deterministic return has positive effect, exposing a positive full-phase/core event;*
- (ii) *it has zero effect, giving mixed-domination deletion or a neutral return handled by exact-forward recurrence; or*

(iii) it has negative effect.

Consequently, on any segment whose deterministic height is already bounded by K , the number of unresolved boundary recurrences is effectively bounded in terms of D, N, K . No uniform multiplicity conclusion is asserted when the entry/peak height is unbounded.

Proof. There are finitely many refined signatures. Equality of two signatures fixes the finite semantic data used by the boundary analysis. The sign of the intervening deterministic return gives the three cases above. In the negative case each recurrence of one fixed signature lowers deterministic height by at least one; under a height bound K this can happen at most $K + 1$ times for that signature. Taking the finite maximum over signatures gives the local bound. $\square$

The opposite difficulty is a large backward requirement with small active forward credit.

Theorem 7.2 (Low-active exact recurrence). *There is an effective constant B_{PB} such that every pure-backward/low-active branch either*

- (i) yields a deletion,
- (ii) yields a positive-return power certificate, or
- (iii) contains at most B_{PB} relevant rejected cuts.

No bound on the absolute magnitude of the backward requirement is needed.

Proof. When all active forward coordinates are bounded, record the exact color

$$\chi(t) = (p_D(t), A_t, H_t|_{A_t}),$$

where A_t is the support of finite backward requirements. This color set is finite. Equality of two colors with nonpositive deterministic effect permits deletion. Equality with positive effect yields a repeatable deterministic return while the active forward envelope does not improve enough to cross the fixed suffix requirement; hence one obtains a power family of counterexamples. $\square$

8 Anchored relative envelopes

Define the deterministic-height anchored envelope

$$F_t(q) = H_t(q) - d_t$$

on reachable coordinates.

Lemma 8.1 (Relative escape). *Let*

$$C_{\text{rel}} = 2(M_2 - 1).$$

If for some cut and state q ,

$$|F_t(q)| > C_{\text{rel}},$$

then a maximizing synchronized prefix path contains a simple pair cycle C with

$$\Delta_N(C) - \Delta_D(C) \neq 0.$$

Proof. If every repeated synchronized pair state had zero relative drift, erase those closed subwalks. The remaining simple pair path has length at most $M_2 - 1$, so the relative displacement is bounded by $2(M_2 - 1)$. $\square$

If no relative-differential core occurs, the anchored phase space becomes finite.

Theorem 8.2 (No-core height bound). *Assume a shortest certificate-free counterexample contains no relative-differential core. Then there is an effective finite anchored color set of size B_F , and*

$$d_t < H_* := B_F^2 + 2B_F + 1$$

at every cut.

Proof. Equal anchored colors at ascent cuts shift the full forward envelope exactly by the deterministic height difference. A positive anchored return either deletes or, by Lemma 5.1, forces a later low deterministic level. Pair each first-hit ascent level with the first later descent through the same level. More than B_F^2 such paired levels force two equal paired anchored colors, and deleting the equal-height positive and negative factors preserves deterministic feasibility and preserves rejection by monotonicity of the fixed middle transformer. This contradicts shortestness. $\square$

Corollary 8.3. *In the no-relative-core branch, shortest counterexamples have an effective length bound*

$$F(H_* - 1, D, N).$$

Thus every remaining unbounded branch repeatedly exposes bounded relative/differential event cores.

9 The reverse-inclusion promise and source–sink slopes

We now exploit the standing promise

$$L(N) \subseteq L(D).$$

Definition 9.1 (Compatible ordered source–sink route). A bounded synchronized source loop X and a later bounded synchronized sink loop Z are *compatible* if there is one fixed factorization of the input and one choice of N control states such that:

- (i) X is a loop at one synchronized D/N pair state and occurs before Z ;
- (ii) Z is a loop at a later synchronized pair state;
- (iii) fixed finite connector paths lead from the initial configuration to X , from X to Z , and from Z to an accepting N suffix;
- (iv) after sufficiently much source credit is supplied, all internal requirements of those fixed connectors are feasible.

A *route class* records only the finite pair-control/SCC and connector types needed for these conditions. There are effectively finitely many route classes once the bounded event universe is fixed.

Lemma 9.2 (Promise source sign). *Let X be an accessible and coaccessible synchronized loop with*

$$\Delta_N(X) = A > 0.$$

If X can be pumped to cross a fixed finite N -suffix requirement, then

$$\Delta_D(X) \geq 0.$$

Proof. If $\Delta_D(X) < 0$, then sufficiently many copies make the selected N run reach the accepting suffix while the deterministic run eventually underflows. This produces a word in $L(N) \setminus L(D)$. $\square$

Suppose an earlier source and later joint sink have effects

$$X : (a, +A), \quad Z : (-b, -B),$$

with $A, B, b > 0$ and, by Lemma 9.2, $a \geq 0$.

Theorem 9.3 (Promise efficiency inequality). *If the source and sink are compatible in one accessible/coaccessible construction, then*

$$aB \geq bA.$$

If $a > 0$, equivalently

$$\frac{A}{a} \leq \frac{B}{b}.$$

Proof. Let δ be the fixed N credit needed by the connector and accepting suffix. For n sink repetitions choose

$$m_n = \left\lceil \frac{nB + \delta}{A} \right\rceil.$$

Then the nondeterministic run is feasible. If $aB < bA$, the deterministic net effect

$$m_n a - nb$$

tends to $-\infty$, so for large n the same word is in $L(N) \setminus L(D)$, contradicting the promise. $\square$

This inequality extends from individual loops to recurrent block cycles in the finite block graphs of bounded source/sink words.

10 Slope potentials and SCC budgets

Fix one realizable source-to-sink route class. Let

$$\rho_+$$

be the maximum positive N -per- D ratio over recurrent source cycles, and let

$$\rho_-$$

be the minimum N -loss-per- D -loss ratio over recurrent sink cycles. By Theorem 9.3,

$$\rho_+ \leq \rho_-.$$

Choose a rational

$$\lambda \in [\rho_+, \rho_-].$$

Write $\lambda = p/q$ in lowest terms and define

$$\Psi = q \Delta_N - p \Delta_D.$$

Theorem 10.1 (Promise slope potential). *Every recurrent block cycle in the compatible source and sink phases has*

$$\Psi \leq 0.$$

Proof. For a source cycle with $\Delta_D > 0$, the inequality is exactly its ratio being at most λ . Source cycles of nonpositive N effect also have $\Psi < 0$. For a sink cycle $(\Delta_D, \Delta_N) = (-d, -n)$ with $d, n > 0$, the inequality is equivalent to $n/d \geq \lambda$. $\square$

The critical zero-slope case is handled by a max-plus potential.

Lemma 10.2 (Critical graph normalization). *Let S be a finite weighted strongly connected graph whose cycle weights are all nonpositive integers. Then there is an integer vertex potential h such that every reduced edge weight*

$$\bar{\psi}(e) = \psi(e) + h(\text{src}(e)) - h(\text{tgt}(e))$$

satisfies

$$\bar{\psi}(e) \leq 0.$$

Hence every reduced edge is either critical, with weight 0, or decreases by at least 1.

Proof. Fix a root and let $h(v)$ be the maximum weight of a root-to- v path. This maximum is finite because there is no positive cycle. The defining inequality for h gives $\bar{\psi}(e) \leq 0$. $\square$

Define the adjusted budget at a mode state r by

$$B_\lambda = qN - pD - h(r).$$

It is preserved on critical edges and decreases by at least 1 on every noncritical edge.

The remaining issue is to bound the budget entering a critical SCC.

Theorem 10.3 (SCC cone separation). *Let S be a synchronized SCC before a later critical sink. Its cycle cone in coordinates (Δ_D, Ψ) contains no vector with*

$$\Delta_D \geq 0, \quad \Psi > 0.$$

Consequently there exists a rational $\mu \geq 0$ such that every cycle C in S satisfies

$$\Psi(C) + \mu \Delta_D(C) \leq 0.$$

Proof. Let K_S be the rational cone generated by the finitely many simple-cycle vectors (Δ_D, Ψ) of S . Suppose that

$$K_S \cap \{(d, s) : d \geq 0, s > 0\} \neq \emptyset.$$

Choose a rational vector in the intersection and clear denominators.

If its deterministic coordinate is strictly positive, multiply the selected cycle multiplicities by a common factor and join their bases by a fixed closed tour inside the SCC. The connector contribution is $O(1)$ while the selected vector scales linearly; for a sufficiently large factor we obtain a closed synchronized macro W with

$$\Delta_D(W) > 0, \quad \Psi(W) > 0.$$

Thus the N-per-D source ratio of W is strictly larger than λ , in contradiction with Theorem 9.3 and the later λ -critical sink.

It remains to justify the boundary case in which the cone vector has deterministic coordinate 0. If S contains any positive-D cycle, add an arbitrarily small positive rational multiple of that cycle; strict positivity of Ψ is preserved, so we reduce to the preceding case. Otherwise every simple cycle of S has nonpositive deterministic effect. A nonnegative combination with total deterministic effect 0 can then use only zero-D cycles. Since its total Ψ is positive, at least one used zero-D simple cycle has positive Ψ ; that cycle itself is a D-neutral/positive-N pumpable source before a genuine negative-D sink, contradicting the promise efficiency inequality (equivalently, its $a = 0$ case).

Hence the two rational cones are disjoint. The rational separating-hyperplane/Farkas alternative gives an effective $\mu \geq 0$ with

$$\Psi(C) + \mu \Delta_D(C) \leq 0$$

for every cycle C in S . $\square$

Corollary 10.4 (Entry-credit bound). *For every path π inside S ,*

$$\Psi(\pi) \leq C_S + \mu D_{\text{in}},$$

for an effectively computable constant C_S . Thus bounded deterministic entry credit implies bounded adjusted budget.

Proof. Erase cycle subwalks. They have nonpositive $\Psi + \mu\Delta_D$ drift. The residual simple path has bounded length and hence bounded drift. $\square$

Proposition 10.5 (Finite-depth funding provenance). *Consider the synchronized product graph and its condensation DAG. If an SCC needs an unbounded amount of deterministic (respectively N) credit that it cannot generate by a cycle-cone direction with nonnegative drift in the other resource, then the required unbounded credit is already present at the entry of that SCC. Iterating this alternative moves strictly backward in the condensation DAG. Hence every unresolved funding chain has depth at most $|Q_D||Q_N|$.*

Proof. For one SCC S , let

$$K_S = \text{cone}\{(\Delta_D(C), \Delta_N(C)) : C \text{ a simple cycle of } S\}.$$

For the D resource, either K_S contains a vector with $\Delta_D > 0$ and $\Delta_N \geq 0$, or rational cone separation supplies a nonnegative linear ranking

$$\eta_D \Delta_D + \eta_N \Delta_N \leq 0, \quad \eta_D > 0, \eta_N \geq 0,$$

on every cycle. In the ranking branch, cycle erasure bounds D growth by a linear function of the N credit and the bounded simple-path transient; if the necessary D credit is unbounded, the corresponding required resource must therefore already be unbounded at SCC entry. The symmetric alternative holds for N.

If both resources are internally required without an unbounded entry seed, the two productivity directions can be combined, after positive rational rescaling, into a cone vector with both coordinates strictly positive. Such a bi-productive SCC contains a bounded synchronized joint-funder macro and is recorded as a source event rather than as unexplained entry credit. Otherwise the unresolved resource is pushed to a predecessor SCC. A condensation-DAG edge is crossed at each push, so the process terminates after at most the number of synchronized product SCCs, bounded by $|Q_D||Q_N|$. $\square$

Consequently, in the cone-separated branch of Corollary 10.4, choose the earliest SCC whose adjusted budget would be unbounded. Proposition 10.5 makes its deterministic entry credit bounded; hence the adjusted budget is bounded as well.

Theorem 10.6 (Joint source–sink reduction). *Consider an ordered compatible source/sink route with*

$$X : (a, +A), \quad Z : (-b, -B), \quad a, A, b, B > 0.$$

Assume the physical D/N endpoint credits of the route lie in a fixed effective bounded box. Then the following dichotomy holds under the promise $L(N) \subseteq L(D)$.

- (i) *If the realized efficiency gap is strict, all recurrent source/sink multiplicities in the localized route are effectively bounded.*
- (ii) *If the realized slope is exact, the noncritical multiplicities are effectively bounded and the critical source/sink relation powers have effective preperiods and periods. Above an effective balanced-period threshold, one source period and one later sink period package can be deleted while rejoining the same separator state: the identical deterministic physical configuration together with the identical critical N relation state seen by the unchanged suffix. Hence a shortest nondecisive exact-slope route also has an effective local multiplicity and literal-length bound.*

Thus at bounded endpoints the route is either an effective bounded local route, an endpoint-preserving shortening, or a decisive local outcome. The later bounded-entry localization theorems are used separately when an endpoint credit escapes; they are not part of this theorem's hypothesis.

Proof. Let ρ_+ be the maximal recurrent source ratio and ρ_- the minimal compatible sink cost. Theorem 9.3 gives $\rho_+ \leq \rho_-$. In the strict case choose $\lambda = \rho_-$ and clear denominators in $\Psi = q\Delta_N - p\Delta_D$. Chronological matching pairs positive deterministic funding only with later negative-D consumption. The finite set of realized type pairs has a positive minimum gap $\varepsilon_{\text{route}}$, so the bounded endpoint/entry budget bounds the total matched funding; unmatched funding is absorbed by the bounded endpoints and simple skeleton. Thus all recurrent multiplicities are bounded.

Suppose now $\rho_+ = \rho_- = p/q$. Normalize by Lemma 10.2. Noncritical edges decrease the adjusted coordinate

$$B_\lambda = qN - pD - h(\text{state})$$

by at least one, so bounded localized endpoints bound their total number. The critical source and sink relations are finite and therefore ultimately periodic. Choose positive relation periods r, s whose deterministic effects balance, say $ra = sb = K$. Because the exact-slope N effects balance as well, deleting these r source periods and the corresponding s later sink periods preserves the net D and critical- N effects. All connectors and internal requirements are bounded, so above an effective protection threshold the middle route stays feasible; after the sink batch the same separator state is rejoined: both the deterministic physical configuration and the critical N relation state are identical to those before deletion. Hence the unchanged suffix is evaluated against exactly the same separator semantics. Such a deletion contradicts shortestness. Below the threshold, the preperiods, periods, noncritical edges, and unmatched endpoint funding range over a finite effective box. This proves the stated local bound. $\square$

11 Drain classification by maximum cycle mean

Let C be a bounded deterministic control-return word. Fix a prefix configuration and a suffix continuation context.

Definition 11.1 (Useful block graph). The *useful C -block graph* has the N control states as vertices. It contains an edge $q \rightarrow r$ if there is an N path over the literal word C from q to r that is relevant to the fixed context: q is reachable from the prefix after some admissible number of C -blocks and r can still reach the accepting suffix after some admissible number of blocks. Give the edge the maximum counter effect among such C -paths and retain also the minimum initial counter requirement of a maximizing path.

For a directed cycle Γ of k block edges, let

$$\text{mean}(\Gamma) = \frac{\text{total } N \text{ counter effect of } \Gamma}{k}.$$

The maximum of these means over useful directed cycles is denoted μ_C ; if the useful graph is acyclic we set $\mu_C = -\infty$.

Because C is bounded, the block graph, all edge effects and requirements, and μ_C are effectively computable. A maximum is attained by a simple block cycle.

Now suppose

$$d = \Delta_D(C) \geq 0$$

and one selected N -mode has negative effect. The selected mode alone is not the right asymptotic invariant; μ_C is.

Theorem 11.2 (Max-cycle-mean trichotomy). *Exactly one of the following holds.*

- (i) $\mu_C < 0$. Then sufficiently long C -powers lose N credit linearly, while D remains feasible; a cofinal $D \setminus N$ power family is obtained.
- (ii) $\mu_C = 0$. The useful block graph has a critical potential as in Lemma 10.2. Only finitely many noncritical edges can occur from a fixed adjusted budget, while critical relation powers are ultimately periodic. This periodicity is not by itself a multiplicity bound. More precisely:
 - (a) if $d = \Delta_D(C) = 0$, recurrence is a zero- D exact-entry problem and is handled by exact-envelope WQO/deletion;
 - (b) if $d > 0$, arbitrarily many critical repetitions are allowed a priori and are classified as deterministic funding. Their multiplicity is not controlled locally. If that funding is later consumed inside a same-level return, it is handled by the chronological exposure/source-sink machinery of Sections 14–15; unmatched terminal growth is handled separately by the level ladder of Section 16.
- (iii) $\mu_C > 0$. A bounded-period positive- N recurrent class exists, so C is not an asymptotic sink; the rejection obstruction transfers to a later suffix demand.

Proof. For $\mu_C < 0$, cycle erasure bounds every length- k block path by $K - \varepsilon k$. For $\mu_C = 0$, Lemma 10.2 bounds the number of noncritical edges because each decreases the adjusted budget by at least one. The remaining critical relation belongs to a finite relation monoid and is therefore ultimately periodic, but an identity relation may repeat arbitrarily often, so no finite-transient conclusion follows from periodicity alone.

If $d = 0$, every C -block returns D to the identical physical configuration. For a concrete exact entry envelope H , the orbit $H, T_C(H), T_C^2(H), \dots$ is effectively computable. Dickson's lemma yields $i < j$ with $T_C^i(H) \leq T_C^j(H)$; in a shortest witness the factor C^{j-i} is then deletable by mixed domination. Thus zero- D recurrence is bounded node-by-node from its exact entry or shortens the witness. If $d > 0$, this deletion need not be available: repeated critical blocks may carry arbitrarily large deterministic credit. We therefore record them as funding and make no local multiplicity claim. When such funding lies inside a later same-level return it must be consumed by negative- D behavior, where Sections 14–15 perform the required chronological matching; if it remains unmatched to the terminal suffix, Section 16 carries it in the last-occurrence level ladder instead of imposing a local multiplicity bound. For $\mu_C > 0$, a maximizing simple cycle gives a positive recurrent mode of period at most $|Q_N|$. $\square$

12 Protecting large sources and localizing relative seed

A subtle point is that a raw positive- N synchronized cycle need not increase the *entire* forward envelope. We therefore work with active projective recurrence.

At a rejected cut let

$$E = \text{supp}(H), \quad A = \{q : R(q) < \infty\}, \quad O = E \cap A.$$

Lemma 12.1 (No forward-only feed into the active suffix). *Let $i < j$ be cuts with the same forward support E and the same finite completion support A , and let W be the intervening factor. No state in $E \setminus A$ can contribute through W to a state of A in a way that can subsequently complete the fixed suffix after j .*

Proof. If $q \in E \setminus A$ had a W -path to some $r \in A$ that could then complete the suffix after j from sufficiently large credit, then q would have a finite completion requirement for the word consisting of W followed by that suffix. Hence $q \in A$ already at cut i , a contradiction. $\square$

Lemma 12.2 (Active forward sign lock). *Suppose two cuts have the same deterministic control state, the same E, A, O , and the same normalized forward profile on O . If the intervening word has deterministic effect e and active scalar drift α , then in a shortest certificate-free counterexample*

$$e > 0 \Rightarrow \alpha > 0, \quad e < 0 \Rightarrow \alpha < 0, \quad e = 0 \text{ is impossible.}$$

Proof. If $e < 0$ and $\alpha \geq 0$, deleting the factor starts the deterministic suffix from a higher counter value. On O the earlier active envelope is no larger than the later one, on $A \setminus E$ the prefix envelope is bottom, and outside A the fixed suffix cannot accept. Hence the deletion preserves rejection. The same argument handles $e = 0, \alpha \geq 0$.

If $e > 0$ and $\alpha \leq 0$, the deterministic side can repeat the return indefinitely. By Lemma 12.1, forward-only states in $E \setminus A$ cannot improve coordinates relevant to the fixed suffix. On O , monotonicity therefore gives a nonincreasing sequence of active envelopes under further copies of the same return. The suffix remains rejected, so a power certificate is obtained. For $e = 0, \alpha < 0$ the same forward contraction gives a neutral pump. Thus only the stated sign choices survive. $\square$

Lemma 12.3 (Active comparable-pair exposure). *Suppose cuts $i < j$ have the same deterministic control state and the same sets E, A, O , their normalized active-forward profiles agree, and*

$$H_j|_O = H_i|_O + \alpha \mathbf{1}_O, \quad e = d_j - d_i > 0, \quad \alpha > 0.$$

Then, in a shortest counterexample, the deterministic suffix after j visits height at most $e - 1$.

Proof. Assume instead that the suffix after j is feasible from the lower height d_i . Delete the factor between i and j . Deterministic acceptance is preserved by assumption. For every $q \in O$,

$$H_i(q) < H_j(q) < R_j(q).$$

For $q \in A \setminus E$ the earlier forward envelope is bottom, and for $q \notin A$ the fixed suffix has no accepting continuation. Hence the deleted word is still rejected by N , contradicting shortestness.

Therefore the suffix is not feasible from height d_i , although it is feasible from $d_j = d_i + e$. Along the original deterministic suffix there is a first point at which lowering all counters by e would make a transition illegal. Immediately before that failure the original counter is at most $e - 1$, as claimed. $\square$

When active spread is bounded, normalized active profiles form a finite set. Therefore unbounded common active scalar produces a positive cycle in a finite weighted active graph.

Theorem 12.4 (Active-source protectability). *In the bounded-active-spread branch, arbitrarily large active common scalar yields a bounded macro P with*

$$\Delta_D(P) > 0, \quad \alpha(P) > 0.$$

Then either:

- (i) *a later bounded negative- D event is exposed, or*
- (ii) *the entry deterministic height is bounded and the anchored relative credit is large, hence Lemma 8.1 exposes a bounded relative-differential core.*

Proof. Finite weighted-cycle extraction gives $\alpha(P) > 0$; Lemma 12.2 gives $\Delta_D(P) > 0$. Lemma 12.3 forces a later deterministic height below the gained credit. From high entry height this descent contains a negative- D simple cycle. From low entry height, the large active scalar becomes large $H - D$, and relative escape applies. $\square$

For the relative branch, use the synchronized SCC cone in coordinates

$$(\Delta_D, \Delta_{\text{rel}}).$$

Theorem 12.5 (Earliest relative source). *Every unbounded positive relative seed can, after descending finitely many SCC-provenance steps, be localized to a bounded-entry synchronized source P satisfying*

$$\Delta_D(P) \geq 0, \quad \Delta_N(P) > 0, \quad \Delta_{\text{rel}}(P) > 0.$$

Both deterministic and selected nondeterministic entry counters are bounded by a computable function of D, N .

Proof. In a synchronized SCC S , put

$$r(C) = \Delta_N(C) - \Delta_D(C)$$

for a cycle C and let

$$K_S^{\text{rel}} = \text{cone}\{(\Delta_D(C), r(C)) : C \text{ a simple cycle of } S\}.$$

If this cone is disjoint from

$$\{(d, r) : d \geq 0, r > 0\},$$

rational separation gives an effective $\mu \geq 0$ such that

$$r(C) + \mu \Delta_D(C) \leq 0$$

on every cycle. Cycle erasure then bounds the relative gain of any path in S by

$$C_S^{\text{rel}} + \mu D_{\text{in}}.$$

Thus bounded deterministic entry credit cannot produce an unbounded positive relative seed inside such an SCC.

Trace an assumed unbounded relative seed backward through the synchronized condensation DAG and use Proposition 10.5 whenever its deterministic entry credit is unbounded. Choose the earliest SCC at which the relative seed is still unresolved. By minimality, both the incoming relative seed and the deterministic entry credit are bounded. Hence the separating branch above is impossible, so K_S^{rel} meets the positive-relative target cone.

Realize a rational target vector by integer simple-cycle multiplicities. If its deterministic coordinate is positive, scale the multiplicities until the bounded connector tour inside the SCC is dominated, producing a closed macro P with $\Delta_D(P) > 0$ and $r(P) > 0$. If the target deterministic coordinate is 0, either perturb by a positive-D cycle and reduce to that case, or every used cycle has deterministic effect 0, in which case one zero-D simple cycle already has positive relative drift. In both cases we obtain

$$\Delta_D(P) \geq 0, \quad \Delta_N(P) = \Delta_D(P) + r(P) > 0.$$

A realization with negative deterministic drift and positive N drift would instead be a pumpable negative-D/positive-N loop and would contradict $L(N) \subseteq L(D)$. Finally, bounded deterministic entry plus bounded incoming relative seed implies bounded selected N entry because $N_{\text{in}} = D_{\text{in}} + r_{\text{in}}$. $\square$

Corollary 12.6 (Quantitative relative-source localization). *For every bounded deterministic entry range $0 \leq D \leq B$, there is a computable threshold*

$$R_{\text{rel}} = R_{\text{rel}}(D, N, B)$$

with the following property. If, at a reachable cut with deterministic counter at most B , some reachable envelope coordinate satisfies

$$H(q) - D > R_{\text{rel}},$$

then the provenance of that coordinate contains an earlier bounded-entry synchronized source P such that

$$\Delta_D(P) \geq 0, \quad \Delta_N(P) > 0, \quad \Delta_{\text{rel}}(P) > 0,$$

and both the deterministic and selected nondeterministic entry counters of P are bounded by a computable function of D, N, B .

Proof. The proof of Theorem 12.5 is effective at every stage. For each synchronized SCC whose cycle cone is disjoint from the positive-relative target cone, rational separation yields an effectively computable bound on relative gain in terms of the deterministic entry credit and the bounded simple-path/connector transient. The synchronized condensation graph has finite depth, and all simple-path, connector, and SCC-separation constants range over effectively finite families. Taking their maximum over all provenance chains reachable with deterministic entry at most B gives a computable threshold $R_{\text{rel}}(D, N, B)$.

Above this threshold the cone-separated alternative cannot persist throughout the provenance chain; hence some SCC must realize a positive-relative cone direction. The constructive part of Theorem 12.5 then yields the stated bounded-entry source P , with both entry counters bounded effectively in terms of D, N, B . $\square$

Pump this source against one fixed exact completion. If many copies are needed, the selected completion contains a low- N point and hence many negative- N simple cycles.

Lemma 12.7 (Bounded-entry first drain). *From the bounded-entry source of Theorem 12.5, either a compact joint negative- D sink branch is already exposed, or one can reach a bounded negative- N core C with both deterministic and selected N entry counters bounded by a computable function of D, N .*

Proof. Let the bounded-entry source be a synchronized loop

$$P : (a, A), \quad a \geq 0, \quad A > 0,$$

based at pair state (p, x) , and let the selected N counter there be $h \leq B_0$. Fix one selected continuation V from x to an accepting state, including the route through the current active cut and an exact completion. Let $\text{req}(V)$ be its minimum required entry credit and set

$$K = \min\{k \geq 0 : h + kA \geq \text{req}(V)\}.$$

If K is uniformly bounded there is no large-seed problem. Otherwise, at the minimal value K the surplus over the required credit is $< A$. Along V choose a prefix ending where the defining deficit of $\text{req}(V)$ is attained. The selected N counter there is at most $A - 1$. Consequently the N -effect of the prefix from the source base to this low point is

$$-(K - 1)A + O(1).$$

Decompose that synchronized pair path into simple cycles plus a simple residual path. The residual path has length at most $M_2 - 1$, so unbounded K forces negative- N simple cycles. Order the removed cycle occurrences chronologically and let C be the first one with negative N effect. Before C , every removed cycle has nonnegative N effect.

If a pre- C cycle had negative deterministic effect and nonnegative N effect, then after supplying the bounded connector requirements one could pump that loop: N would not lose

credit, while D would eventually underflow. This contradicts the promise $L(N) \subseteq L(D)$. Hence every recurrent piece before the first negative-N cycle has nonnegative deterministic effect. Delete those pre- C recurrent pieces and retain only the bounded simple product skeleton. Doing so cannot create an unbounded D- or N-credit requirement. Since the skeleton length and all primitive requirements are bounded, the base of the same finite negative-N cycle type C has a realization with both D and selected N counters bounded by a computable function of D, N , unless a negative-D/negative-N joint sink has already been exposed. $\square$

13 Bounded-entry deferral and scalar compactness

Apply Theorem 11.2 to the bounded-entry first drain C .

If $\Delta_D(C) < 0$, we are in the joint sink branch controlled by the slope potential and SCC budget. If $\mu_C < 0$, the drain either becomes decisive or is left after an effective bounded number of blocks. If $\mu_C = 0$ and $\Delta_D(C) = 0$, this is a neutral exact-entry recurrence and is deferred to the WQO/deletion treatment; no uniform relation-monoid transient is asserted. If $\mu_C = 0$ and $\Delta_D(C) > 0$, the branch is not a bounded local transient at all: it is recorded as a positive-D, zero-mean funding segment and passed forward to the later exposure/source-sink analysis, or to the terminal level ladder if the funding remains unmatched. Only $\mu_C > 0$ creates the positive-mean deferral chain treated below.

Theorem 13.1 (Uniform deferral entry bound). *There is a total computable bound $B_{\text{src}}(D, N)$ such that every source in a surviving chain of positive-mean deferrals is entered with*

$$D \leq B_{\text{src}}, \quad N \leq B_{\text{src}}.$$

Moreover there is a total computable local peak bound $B_{\text{episode}}(D, N)$ valid throughout each source-to-drain-to-source episode.

Proof. Fix a bounded first-drain word C entered with counters at most B . Its useful block graph has finitely many control states, bounded edge effects, and bounded internal requirements.

If $\mu_C < 0$, choose an effective $\varepsilon_C > 0$ and K_C such that every useful block path of k blocks has N effect at most $K_C - \varepsilon_C k$. Starting from credit at most B , the route must either reject or leave this negative-mean class within an effectively bounded number of blocks.

The zero-mean cases are not used to manufacture a deferral-entry bound. When $\Delta_D(C) = 0$ they are neutral exact-entry recurrences, and when $\Delta_D(C) > 0$ they are deterministic-funding segments whose repetition may be unbounded until later negative-D exposure. Both cases leave the positive-mean deferral chain rather than being declared finite transients.

If $\mu_C > 0$, choose a positive critical SCC. From the bounded entry, either that SCC is physically unreachable—in which case it is removed from the useful graph and the true reachable maximum mean is nonpositive—or it is reached by a simple block-SCC path plus only finitely many repetitions needed to meet the bounded internal requirements. Both the access length and the counter value at first entry are therefore bounded effectively by the finite type of C and by B . The first positive critical cycle reached there is the next source.

The first-drain construction preceding this classification already supplies one D,N-only bound on the entry counters of every drain type. Taking the maximum, over the finite bounded core universe, of (i) those drain-entry bounds and (ii) all positive-critical-class first-entry bounds yields a single number B_{src} . Importantly, the next source is reached through the bounded drain-entry realization just described; its entry bound does *not* depend on the possibly large peak credit generated in the conceptual source-pumping argument. Thus the same maximum applies at every deferral. Taking also the maximum counter excursion along the finitely many bounded access fragments yields B_{episode} . $\square$

Corollary 13.2 (Relative-seed compactness). *No unbounded positive relative seed is transported through the surviving positive-mean deferral chain. Every retained deferral episode is representable by finite event data together with entry/exit scalar values from a bounded set depending only on D, N . This corollary makes no bounded-height claim for the separate zero-mean positive- D funding segments.*

14 Full-envelope differential control in bounded-return excursions

The preceding source/seed analysis can be carried out on selected synchronized runs, but the global proof must control the *entire* nondeterministic forward envelope. We now establish the required all-mode statement.

Choose once and for all a computable integer B_0 larger than every bounded primitive deterministic effect, every bounded internal deterministic requirement used in the event/core decomposition, and the bounded source/episode constants produced above. A cut is called *LOW* when its deterministic counter value is at most B_0 . A *same-level bounded-return excursion* begins at deterministic height $h > B_0$, never goes below h , and returns to h .

14.1 Triple-product differential geometry

Consider the synchronized graph

$$G_3 = Q_D \times Q_N \times Q_N.$$

A triple edge follows the unique deterministic transition and two nondeterministic strands over the same input symbol. Give it differential weight

$$g = \Delta_N(\pi) - \Delta_N(\rho).$$

Lemma 14.1 (One-sign differential potential). *Let S be an SCC of G_3 . If S has no negative g -cycle, then there is a computable potential ϕ and a constant C_S such that every path inside S changes the strand gap by at least $-C_S$. Dually, if S has no positive g -cycle, every path changes the gap by at most C_S . If every cycle has $g = 0$, then g is an exact endpoint-potential difference.*

Proof. In the no-negative case, fix a root and let $\phi(v)$ be the minimum g -weight of a root-to- v path. The minimum is finite because there is no negative cycle, and a minimizing path may be taken simple. Hence for every edge $u \rightarrow v$,

$$g(e) \geq \phi(v) - \phi(u).$$

Summing along a path gives the claim. Apply the same argument to $-g$ for the dual statement. If both signs are excluded, the inequalities force equality on every cycle and hence exact potential behavior. $\square$

Lemma 14.2 (Mixed-sign zero-differential package). *If a triple SCC contains a positive and a negative differential cycle, then it contains an effectively bounded same-base closed macro Z on which the two strands have equal total N effect.*

Proof. Join a common base to bounded simple positive and negative differential cycles and back by simple connectors. Repeat the positive and negative cycles only enough to dominate the bounded connector error, obtaining same-base macros of differential effects $A > 0$ and $-B < 0$. Taking $B/\gcd(A, B)$ copies of the first and $A/\gcd(A, B)$ copies of the second gives total differential effect zero. $\square$

Fix a total order on N transitions. At every suffix-relevant envelope update, if several predecessor runs attain the same maximum, choose the least one in this order as the *canonical owner*. When the canonical owner changes from one maximizing strand to another, follow the two competing strands in G_3 . Inside a fixed triple SCC S , define the *switch signature* to consist of

(triple vertex, ordered old/new owner states, γ , event/regime label),

where γ is the integer height gap between the two competing strands at the switch cut. Boundary and relative events are recorded separately rather than included as ordinary signatures.

Theorem 14.3 (Owner stabilization outside mixed-sign SCCs). *Arbitrarily many semantically relevant envelope-owner changes in a bounded-return excursion can occur only inside a mixed-sign triple SCC. In a one-sign or all-zero SCC, either the number of owner switches is effectively bounded, or a repeated switch signature exposes an effectively bounded same-base zero-differential package. The only other exits are an already recorded boundary event or relative event.*

Proof. Let S first have no negative differential cycle; the dual case is symmetric. By Lemma 14.1, along every path in S the strand gap can decrease by at most the computable potential oscillation

$$C_S = \max_S \phi - \min_S \phi.$$

Hence if one candidate leads another by more than C_S (enlarged by the finite one-edge/event correction), the trailing candidate cannot become the canonical maximum before the path exits S . Every owner switch inside S therefore occurs with γ in one fixed finite integer interval. The triple vertex, ordered owner pair, and event/regime label also range over finite sets, so there are only finitely many switch signatures.

If more switches occur than signatures, choose two occurrences with the same signature. The intervening synchronized triple walk begins and ends at the same triple vertex and the same gap value, hence has total differential weight 0. In a one-sign SCC every directed simple cycle has differential weight of the same weak sign. Simple-cycle erasure of this closed walk therefore contains a simple cycle of differential weight 0; its length is at most $|Q_D||Q_N|^2$. Together with the bounded connectors to its base this is the required bounded same-base zero-differential package. If every cycle of S already has weight 0, the same conclusion is immediate from the exact endpoint-potential case of Lemma 14.1.

The condensation graph of G_3 is acyclic, so an excursion can change triple SCC only finitely many times. A switch that leaves the stable support/requirement regime is, by definition, a boundary or relative event. Thus unbounded switching without one of the stated bounded packages can persist only in a mixed-sign SCC, where Lemma 14.2 supplies the separate mixed-sign reduction. $\square$

14.2 Positive relative escape and lag/catch-up reduction

The positive-relative branch admits a full-envelope localization, not merely a selected-strand one.

Theorem 14.4 (Canonical positive-relative source cut). *For every bounded deterministic entry range there are computable constants B_{src} and B_{full} with the following property. If at a bounded- D cut some reachable coordinate satisfies $H(q) - D$ above the corresponding positive-relative threshold, then at an earlier cut s there is a bounded positive-relative source and*

$$D(s) \leq B_{\text{src}}, \quad \max_q H_s(q) \leq B_{\text{full}}.$$

Proof. Fix an a priori deterministic entry bound B for the cut in the theorem. Corollary 12.6 supplies the effective threshold $R_{\text{rel}}(D, N, B)$ and, whenever that threshold is crossed, a computable

bound on the deterministic and selected-N entry counters of the resulting localized source. Let B_{src} dominate those source-entry bounds over the finite source/regime types. Let C_* be the maximum bounded connector/simple-path correction occurring in the finite source-provenance construction, and choose one computable threshold R^* satisfying

$$R^* \geq R_{\text{rel}}(D, N, B), \quad R^* \geq R_{\text{rel}}(D, N, B_{\text{src}}) + C_*.$$

For every envelope coordinate whose anchored value $H(q) - D$ exceeds R^* at the given cut, apply Corollary 12.6 to a realizing predecessor strand. This produces a bounded-entry positive-relative source certificate in the prefix. Among all such certificates feeding all large coordinates, choose one whose *entry cut* s is leftmost; ties at the same cut are broken by the fixed finite order on source types. By construction,

$$D(s) \leq B_{\text{src}},$$

and the selected N entry coordinate of that certificate is bounded as well.

We claim that every other reachable coordinate at the same entry cut is bounded. Suppose not. Then for some r ,

$$H_s(r) > B_{\text{src}} + R^*.$$

Since $D(s) \leq B_{\text{src}}$, we have

$$H_s(r) - D(s) > R^* \geq R_{\text{rel}}(D, N, B_{\text{src}}) + C_*.$$

After absorbing the bounded connector/simple-path correction, the anchored seed still exceeds the quantitative threshold for deterministic entry range B_{src} . Apply Corollary 12.6 to a realizing strand for $H_s(r)$, now using the prefix ending *at* s . The large seed is already present before the chosen source macro begins, so its provenance certificate has an entry cut strictly to the left of s ; a certificate tied at s cannot generate credit that is already present at its own entry. This contradicts the leftmost choice. The same argument covers ties among several maximum coordinates at the original cut because the leftmost minimization is taken simultaneously over all coordinates exceeding R^* .

Thus every finite $H_s(r)$ lies below the computable common bound

$$B_{\text{full}} := B_{\text{src}} + R^*.$$

Taking the maximum over the finite source/regime types gives one pair of constants $B_{\text{src}}, B_{\text{full}}$ for the stated bounded deterministic entry range. $\square$

Call such a bounded full-envelope cut *POS-canonical*. Between POS-canonical cuts, any remaining unbounded spread is negative relative lag. Follow a predecessor chain that later becomes semantically relevant. A large catch-up contains a bounded positive-relative core. Outside canonical regeneration its deterministic effect must be negative; if its N effect were positive, pumping it would eventually accept in N while underflowing D , contradicting the promise. Thus surviving catch-up is N-nonpositive. Dually, a genuinely new lag generator has positive deterministic effect and negative relative effect.

Proposition 14.5 (Lag/catch-up sign reduction). *Outside already classified drain/demand branches and POS-canonical regeneration, every recurrent lag/catch-up pair has the form*

$$C_{\text{lag}} : (+a, -A), \quad C_{\text{up}} : (-b, +B),$$

with $a, A, b, B > 0$ and $\Delta_N(C_{\text{up}}) = -b + B \leq 0$. Any positive-N recurrent mode on the lag side reduces to the strict/exact source-sink program or contradicts the standing promise. Hence the only unresolved recurrent pair has nonpositive useful N maximum mean on both sides, with zero-mean residues recorded by finite critical relation states.

Proof. Write a recurrent macro in coordinates (d, r) , where $d = \Delta_D$ and $r = \Delta_N - \Delta_D$ is the anchored relative effect.

A genuine new lag generator has $r < 0$. If $d < 0$, it is already a negative-D drain/demand event; if $d = 0$, it is a zero-D losing recurrence handled by the neutral/mean classification. Thus an unresolved lag generator must have $d = a > 0$ and $r = -A < 0$, giving $C_{\text{lag}} : (+a, -A)$.

A catch-up macro has $r > 0$. If $d \geq 0$, repetition creates an unbounded positive relative seed. Theorem 12.5 first localizes that seed to bounded deterministic entry, and Theorem 14.4 then regenerates a POS-canonical cut. Outside that branch we must therefore have $d = -b < 0$. Write $r = B > 0$. Its N effect is $-b + B$. If this were positive, the recurrent macro could be pumped: N would gain credit while D loses credit, eventually producing an N-accepting/D-underflowing word and contradicting $L(N) \subseteq L(D)$. Hence $-b + B \leq 0$.

Finally consider any other recurrent useful mode on the lag side. A mode with positive N effect and negative D effect contradicts the promise as above; with zero D effect its repetition creates a positive-relative seed and triggers bounded-entry localization followed by canonical regeneration; with positive D effect it is a genuine source whose later negative-D consumption is governed by the strict/exact source-sink alternatives. After removing these cases, no surviving useful block graph can have positive maximum N mean. Zero-mean critical behavior is retained only through its finite critical relation state; no multiplicity bound is inferred from relation periodicity itself. $\square$

14.3 A combined useful-mode potential

Individual nonpositive means are not enough: alternating different event words could create a positive cycle. We first perform an explicit positive-cycle reduction and only then apply a max-plus potential.

Let M_{raw} be the finite graph obtained from all surviving bounded event modes after the owner and lag/catch-up reductions. A vertex records the deterministic boundary control, event/core type, finite critical residue, N state, support/coaccessibility regime, and low/high requirement regime. Each useful macro edge carries deterministic effect $d(e)$ and N effect $n(e)$; internal requirements are absorbed into the finite regime label.

Lemma 14.6 (Zero-D closed-mode compactness). *Fix a concrete exact entry envelope H . Let $\mathcal{Z}$ be the finite set of primitive simple closed mode packages in M_{raw} whose net deterministic effect is zero, together with the bounded simple zero-D connectors between their base vertices. Arbitrary recurrence generated by $\mathcal{Z}$ has an effective node-dependent bound in a shortest witness, unless an endpoint-preserving deletion has already occurred.*

Proof. A state of the local search is the finite combined-mode vertex together with the full exact forward envelope at a cut where D is in the same physical configuration as at the corresponding package base. The package/connector set is finite, and every exact envelope successor is computable, so the search is effectively finitely branching. If along one branch two states have the same finite mode vertex and envelopes $H_i \leq H_j$ with $i < j$, the intervening factor returns D to the identical physical configuration; mixed-domination deletion therefore preserves D acceptance and N rejection of the unchanged suffix. A shortest witness consequently gives a bad sequence in a finite disjoint union of copies of $\mathbb{N}^{|Q_N|}$. Dickson excludes infinite branches and König makes the finitely branching search tree finite. Its finite maximum depth and maximum encountered envelope give the claimed computable exact-entry bounds. Longer zero-D closed walks cause no extra case: simple-cycle erasure decomposes them into members of $\mathcal{Z}$ plus bounded simple connectors. $\square$

Lemma 14.7 (Chronological compatible matching in a same-level return). *Consider the synchronized event trace of one same-level bounded-return excursion. After cycle erasure, positive-D recurrent occurrences can be greedily matched, unit by unit and in chronological order, to later*

negative-D recurrent consumption, except for a computably bounded amount absorbed by the residual simple skeleton and the two bounded endpoint fragments. Moreover every matched source/sink pair belongs to one of finitely many compatible ordered route classes of Section 9.

Proof. Decompose the synchronized trace into its residual simple pair-control skeleton and its chronologically located recurrent cycle occurrences. Give each positive-D occurrence $d > 0$ funding tokens and each negative-D occurrence $-d$ consumption slots. Scan from left to right, placing source tokens in a queue and assigning each negative-D slot to the earliest available earlier token. Because the complete excursion returns to its initial deterministic level, every token not matched to a recurrent negative-D occurrence must be consumed by the final residual/end fragment; similarly an initial deficit can only be supplied by the initial residual/end fragment. The residual pair-control path is simple after cycle erasure and the endpoint fragments come from a finite bounded universe, so the total unmatched correction is computably bounded.

For a matched source occurrence X and later sink occurrence Z , keep their synchronized pair states and erase recurrent subwalks only from the factor strictly between them. The remaining connector is simple in the finite synchronized pair graph and therefore bounded. Accessibility is inherited from the prefix of the original trace and coaccessibility from its suffix. Internal requirements, support/coaccessibility regime, and critical residue range over the finite event data. Hence the matched pair is assigned to one of the finitely many compatible ordered source–sink route classes defined in Section 9, with chronology preserved. If its endpoint credit is bounded, Theorem 10.6 applies to that class; if an endpoint credit escapes, the bounded-entry localization is invoked before any multiplicity conclusion is drawn. $\square$

Lemma 14.8 (Positive-cycle elimination). *Fix a concrete same-level bounded-return excursion and exact entry envelope H_0 . Every useful positive-N directed cycle of M_{raw} is removed from the unresolved recurrent part by one of the following mechanisms:*

- (i) *if $d(\Gamma) < 0$, repetition contradicts $L(N) \subseteq L(D)$;*
- (ii) *if $d(\Gamma) = 0$, it belongs to the zero-D closed-mode system of Lemma 14.6, hence has a bounded exact-entry WQO transient or yields deletion;*
- (iii) *if $d(\Gamma) > 0$, its deterministic funding is chronologically matched to later negative-D consumption because the excursion returns to the same level. A useful sink with nonnegative N effect contradicts the promise. A negative-N sink is handled by the strict/exact alternatives of Theorem 10.6; if endpoint credit escapes, the existing bounded-entry localization replaces the escape by a source–drain/deferral transition whose entry and exit scalar values lie in one fixed computable bounded set.*

No claim is made that the number of bounded-entry deferral transitions is itself bounded.

Proof. Decompose a positive directed cycle into directed simple cycles. Since the total N weight is positive, at least one simple component has positive N weight. If such a component has negative D effect, case (i) applies. If it has positive D effect, it is a genuine recurrent source and chronological matching in the same-level return gives case (iii). If every positive-N simple component has zero D effect, the positive part of the recurrence lies in the finite zero-D package system of Lemma 14.6, giving case (ii). Thus the three cases are exhaustive.

For case (iii), cycle erasure of the whole same-level return leaves a bounded simple skeleton and pairs recurrent positive D units, up to bounded unmatched correction, with later negative-D consumption. The source–sink efficiency theorem and Theorem 10.6 handle bounded-endpoint strict and exact pairs. If an endpoint credit escapes, the bounded-entry localization machinery first replaces it by the source–drain/deferral transition described above. When that localization transfers to another positive source, the uniform deferral theorem gives one fixed bound on the source-entry scalar values and one fixed bound on the internal scalar peak of each transfer;

it does *not* assert a bound on the number of transfers. We therefore keep such transfers as finite-state edges rather than summing their number. $\square$

For the fixed exact entry (p, H_0) construct the reduced graph $M_{\text{red}} = M_{\text{red}}(D, N, p, H_0)$ by the following explicit operation.

- (R1) Remove branches already discharged by a promise contradiction or an endpoint-preserving deletion.
- (R2) Replace every zero-D closed-mode episode admitted by Lemma 14.6 by one of the finitely many exact entry-to-exit macro edges enumerated by its finite WQO search tree.
- (R3) Replace every bounded-endpoint strict/exact source–sink episode below its shortening threshold by an exact endpoint macro edge; balanced-period shortening branches are removed rather than retained.
- (R4) Retain bounded-entry deferral transfers as ordinary edges. Split such edges by their exact bounded D/N scalar entry and exit values, synchronized control/event type, support/coaccessibility label, critical residue, and requirement regime. Thus a retained deferral edge records an exact scalar transfer, not merely an upper bound.
- (R5) Every reduced edge records its net deterministic effect $d(e)$, exact tracked N effect $n(e)$, a computable literal-length bound, and a computable internal-peak correction; finitely many endpoint/representative alternatives are split into separate edges.

The scalar ranges and primitive type universes in (R2)–(R5) are finite, so M_{red} is finite even though a path may traverse retained deferral edges arbitrarily many times. No reduction step asserts that the number of such deferral transfers is bounded.

Lemma 14.9 (No positive cycle in the reduced graph). *Every directed cycle Γ of M_{red} satisfies*

$$n(\Gamma) \leq 0.$$

Proof. Because retained deferral edges carry exact scalar entry/exit labels, a directed cycle consisting only of such transfers telescopes in the tracked N scalar and has net N effect zero. More generally, expand every reduced macro edge of a directed cycle to the exact bounded representative specified by (R2)–(R5). If the resulting closed useful walk had positive tracked N effect, simple-cycle decomposition of that expanded walk would expose a positive useful recurrent cycle of M_{raw} . Lemma 14.8 then sends that recurrence to contradiction, zero-D WQO compactness, bounded strict/exact source–sink reduction, or retained bounded-entry deferral. The first three outcomes are not recurrent edges of M_{red} , while the last has already been split by its exact scalar transfer and cannot create a positive closed scalar gain. Hence every directed cycle of M_{red} has nonpositive N weight. Notice that the raw graph itself is not claimed to have this property. $\square$

Theorem 14.10 (Arbitrary-interleaving envelope bound). *Fix a same-level bounded-return excursion entered with deterministic control p and exact forward envelope H_0 . There is a computable constant $C_{\text{BR}}(D, N, p, H_0)$ such that every suffix-relevant forward-envelope coordinate throughout any retained unresolved excursion is at most*

$$\text{maxfin}(H_0) + C_{\text{BR}}(D, N, p, H_0).$$

The constant includes the finite internal-peak corrections of the eliminated zero-D and source–sink episodes.

Proof. Inside each relevant SCC of M_{red} define $h(v)$ as the maximum N weight of a fixed SCC root-to- v path. Lemma 14.9 makes this maximum finite and gives

$$n(e) + h(u) - h(v) \leq 0$$

on every intra-SCC reduced edge. The condensation graph is acyclic, so an arbitrary path crosses at most finitely many inter-SCC edges; the maximum total positive contribution of those simple condensation-DAG transitions is an effective constant C_{DAG} . Hence the reduced part of an arbitrary interleaving increases the tracked N credit by at most $\max h - \min h + C_{\text{DAG}}$, independently of how often bounded-entry deferral edges are traversed inside SCCs. Every collapsed zero-D or source-sink edge has a computable internal peak above the larger of its endpoint credits, and the uniform deferral theorem gives the same kind of fixed peak bound for each retained deferral edge. Taking the maximum of these *per-edge peak* corrections over the finite reduced graph, rather than multiplying by the number of episodes, and adding the primitive connector correction gives $C_{\text{BR}}(D, N, p, H_0)$. $\square$

With the full envelope bounded, canonical event cuts carry finitely many exact active colors

$$\chi = (p_D, A, H|_A),$$

where A is the finite completion support.

Lemma 14.11 (Exact active-color deletion). *Let $i < j$ be canonical cuts of a rejected word with the same exact active color*

$$(p_D, A, H|_A)$$

and suppose the deterministic effect of the intervening factor is nonpositive. Then deleting that factor preserves deterministic acceptance and nondeterministic rejection of the unchanged suffix. Consequently such a repetition cannot occur in a shortest counterexample.

Proof. Equality of colors gives the same deterministic control state at the two cuts, the same completion support A for the suffix after j , and

$$H_i|_A = H_j|_A.$$

A nonpositive deterministic effect means that the deterministic counter at i is at least its value at j . Therefore the suffix accepted by D from cut j remains executable and accepting from cut i by monotonicity.

Because the original word is rejected by N , every $q \in A$ satisfies $H_j(q) < R_j(q)$. The exact-color equality gives the same inequality with $H_i(q)$. States outside A have no accepting continuation along the suffix after j . Hence the suffix after j is still rejected from the envelope at cut i . Deleting the interval therefore produces a shorter word in $L(D) \setminus L(N)$, a contradiction. $\square$

A positive-effect repetition is treated in the next section by deterministic funding rather than by appending the inherited suffix to a power certificate.

15 Context-safe local recurrence

The structural analysis above exposes several local recurrence families. The global proof must not assume that a short local seed implies a short inherited suffix. We now show that every such family is context safe.

15.1 Neutral returns

Lemma 15.1 (Exact-entry neutral power cap). *Let x be a fixed word that returns D to the identical physical configuration. From a concrete exact forward envelope H , define $H_t = T_x^t(H)$. There is a computable $J_x(D, N, H)$ such that a shortest $D \setminus N$ witness contains fewer than J_x consecutive copies of x from that entry.*

Proof. Encode bottom and finite heights in $\mathbb{N}^{|Q_N|}$. By Dickson's lemma the computable orbit $H_0, H_1, \dots$ has $i < j$ with $H_i \leq H_j$, and the least such pair is found by enumeration. At the two cuts D has the identical physical configuration, so mixed-domination deletion removes x^{j-i} and preserves rejection of the unchanged suffix, contradicting shortestness. $\square$

Consequently every fixed bounded-expression neutral family is bounded sequentially from its exact entry envelopes: bound the first neutral exponent, propagate through the bounded connector, bound the second, and so on. No backward requirement magnitude is used.

15.2 Positive-return color cycles

The preceding full-envelope theorem gives a finite exact active-color universe before any positive-return multiplicity bound is used. We exploit the deterministic balance of a same-level return directly, avoiding any inference from bounded instantaneous N credit to bounded cumulative N loss.

Put

$$B_H = \text{maxfin}(H_0) + C_{\text{BR}}(D, N, p, H_0).$$

Refine every reduced vertex at a canonical event cut by its exact active color

$$\chi = (p_D, A, H|_A).$$

Because $H|_A$ is bounded by B_H , the resulting exact-color graph $G_{\text{col}}(D, N, p, H_0)$ is finite. The reduction (R1)–(R5) gives, before this refinement, a finite set of bounded local edge representatives. After splitting them by their exact color endpoints, let

$$L_{\text{edge}} = L_{\text{edge}}(D, N, p, H_0), \quad D_{\text{edge}} = D_{\text{edge}}(D, N, p, H_0)$$

be respectively the maximum literal length and maximum absolute deterministic effect of one such edge. These maxima are computable from the concrete exact entry and do not use the number of color cycles, the terminal ladder, the LOW tree, or the final hard bound.

Lemma 15.2 (Positive deterministic sign of a retained color cycle). *Every directed cycle occurring in the exact-color trace of a shortest retained same-level bounded-return excursion has strictly positive net deterministic effect.*

Proof. A directed color cycle starts and ends with the same deterministic control, completion support, and exact active envelope. If its deterministic effect were nonpositive, Lemma 14.11 would remove the whole interval while preserving D feasibility and N rejection of the unchanged suffix. That contradicts shortestness. Since deterministic effects are integral, every retained color cycle therefore has effect at least 1. $\square$

Theorem 15.3 (Positive-return total multiplicity bound). *For every concrete exact same-level entry (p, H_0) , the total number of canonical event edges, and hence the total number of positive-return exact-color recurrences, in a shortest retained bounded-return excursion is effectively bounded under arbitrary interleaving.*

Proof. Apply ordinary cycle erasure to the finite vertex sequence of G_{col} traced by the excursion. It decomposes the trace into a simple residual path σ and a multiset of directed simple color cycles $\Gamma_1, \dots, \Gamma_m$. Write

$$C_{\text{col}} = |V(G_{\text{col}})|.$$

The simple path uses at most $C_{\text{col}} - 1$ edges, so

$$d(\sigma) \geq -(C_{\text{col}} - 1)D_{\text{edge}}.$$

The complete same-level excursion has deterministic net effect zero (bounded entry/exit fragments may be included as distinguished color edges; if they are kept outside the trace, their finite D correction is simply added to the right-hand side). Hence

$$\sum_{j=1}^m d(\Gamma_j) = -d(\sigma) \leq (C_{\text{col}} - 1)D_{\text{edge}} + C_{\text{end}},$$

where C_{end} is the computable bounded end-fragment correction. By Lemma 15.2, every summand is a positive integer. Therefore

$$m \leq (C_{\text{col}} - 1)D_{\text{edge}} + C_{\text{end}}.$$

Each simple cycle has at most C_{col} edges. Consequently the entire canonical event trace has at most

$$K_{\text{evt}} := (C_{\text{col}} - 1) + C_{\text{col}}((C_{\text{col}} - 1)D_{\text{edge}} + C_{\text{end}})$$

edges. Every positive-return exact-color recurrence is contained in this finite trace, so its total multiplicity is bounded as well.

The proof uses only deterministic balance, exact-color deletion, and bounded local edge effects. In particular, repeated reduced modes may replenish N credit arbitrarily within the already bounded local edges; no cumulative N-loss estimate is used or needed. $\square$

15.3 Negative and zero mean

Let C be a bounded recurrent core with $d = \Delta_D(C) \geq 0$ and concrete exact entry envelope H . Fix only the finite completion support A of the inherited suffix.

If the useful C -block graph has maximum mean $\mu_C < 0$, cycle erasure computes $\varepsilon_C > 0$ and K_C such that every useful k -block path has N effect at most $K_C - \varepsilon_C k$. Therefore there is a support-parametric kill threshold

$$K_{\text{kill}} = 1 + \left\lceil \frac{\max_{\text{fin}}(H) + K_C}{\varepsilon_C} \right\rceil$$

(up to the finite internal-requirement correction) after which no useful route can reach A with nonnegative credit. If $d = 0$, excess copies delete. If $d > 0$, an excess copy either deletes or is indispensable only because its deterministic funding is consumed later, forcing a later bounded-D exposure and transfer to the LOW/source-sink machinery.

If $\mu_C = 0$, a critical potential bounds all gain relative to the exact entry. Suffix interaction reduces to a finite semantic quotient: either a large-deficit class, where no power can reach the completion support, or one of finitely many bounded-deficit critical relation states. Periodic recurrence at $d = 0$ deletes directly. At $d > 0$, failure of deletion again exposes later deterministic consumption.

Theorem 15.4 (Mean-recurrence context elimination). *Negative-mean and zero-mean local recurrences from a concrete exact entry contribute only a bounded local transient, endpoint-preserving deletion, deterministic-funding transfer to a later bounded-D exposure, or ordinary continuation. No complete witness bound is obtained by appending an inherited suffix.*

15.4 Strict and exact source–sink routes

For a compatible positive-D source C_+ and negative-D sink C_- define

$$\rho(C_+) = \frac{\Delta_N(C_+)}{\Delta_D(C_+)}, \quad \kappa(C_-) = \frac{-\Delta_N(C_-)}{-\Delta_D(C_-)}.$$

The promise gives $\rho \leq \kappa$. If endpoints are bounded and the inequality is strict, chronological matching charges each unit of deterministic funding an N-budget loss of at least one positive computable gap $\varepsilon_{\text{route}}$. Hence total matched funding, and thus route length, is bounded. If an endpoint N credit escapes while D is bounded, anchored relative escape localizes it to a bounded-entry source/drain episode, which returns to the already classified mean/source–sink cases rather than carrying an unbounded suffix scalar.

At exact slope $\rho = \kappa = p/q$, use the adjusted coordinate

$$B_\lambda = qN - pD - h(\text{state}).$$

Noncritical edges decrease it and critical relation powers have finite preperiods and periods. Above an effective protection threshold, balanced source and sink critical periods can be deleted together so that the same separator state — the same D physical configuration and the same critical N relation state — is rejoined before the unchanged suffix.

Theorem 15.5 (Source–sink context elimination). *Every source–sink route is reduced, without bounding the inherited suffix, to one of:*

- (i) *a bounded strict local route;*
- (ii) *a bounded exact-critical local route;*
- (iii) *endpoint-preserving balanced-period shortening;*
- (iv) *bounded-entry localization to an already classified local mechanism; or*
- (v) *a contradiction to $L(N) \subseteq L(D)$.*

15.5 The exact-entry bounded-return theorem

The proof of a whole-excursion bound must not merely take the maximum of per-type transients, because finitely many event types could alternate arbitrarily often. Theorem 14.10 first gives the explicit bound

$$B_H(D, N, p, H) := \text{maxfin}(H) + C_{\text{BR}}(D, N, p, H)$$

on every suffix-relevant exact forward-envelope coordinate at canonical cuts. After B_H is known, all support/regime data and exact active envelopes range over finite sets. Thus the bounded local representatives produced by (R1)–(R5) can be split by exact color endpoints and one can compute the finite exact-color graph G_{col} together with L_{edge} and D_{edge} . Theorem 15.3 then gives the explicit event-edge bound K_{evt} by deterministic cycle erasure. Only after these quantities are available do we take maxima of the local literal edge lengths. Hence the construction never infers a whole-factor bound from “finitely many schema types” alone.

Theorem 15.6 (Exact-entry bounded-return compactness). *For every concrete same-level high entry consisting of deterministic control p and exact forward envelope H , there is a total computable integer*

$$L_{\text{BR}}^\dagger(D, N, p, H)$$

such that every same-level bounded-return factor occurring in a shortest hard-direction counterexample has length at most $L_{\text{BR}}^\dagger$ unless the local analysis already yields endpoint-preserving shortening, a standing-promise contradiction, or a bounded decisive continuation from the current exact node.

Proof. Compute successively the zero-D closed-mode tree of Lemma 14.6, the positive-cycle elimination corrections, the reduced-graph potential constant C_{BR} , the full-envelope bound B_H , the finite exact-color graph G_{col} with its local constants $L_{\text{edge}}, D_{\text{edge}}$, and finally the deterministic event budget K_{evt} of Theorem 15.3. This dependency is acyclic: none of these quantities uses $L_{\text{BR}}^\dagger$, the terminal ladder, the LOW successor tree, or the final hard bound. Since every interval between consecutive canonical cuts has literal length at most L_{edge} , multiplying K_{evt} by L_{edge} and adding the bounded end fragments yields a total computable $L_{\text{BR}}^\dagger(D, N, p, H)$. $\square$

The first two exceptional outcomes cannot survive on the retained shortest promised branch. A bounded decisive continuation is charged with the already traversed prefix in the global accounting below.

Positive-return dependency re-audit. The construction order is now explicit:

$$C_{\text{BR}} \longrightarrow B_H \longrightarrow (G_{\text{col}}, L_{\text{edge}}, D_{\text{edge}}) \longrightarrow K_{\text{evt}} \longrightarrow L_{\text{BR}}^\dagger.$$

The event budget K_{evt} is obtained from deterministic cycle erasure in the already finite exact-color graph. It does not use a cumulative N-loss bound and does not feed back into C_{BR} or B_H .

Dependency checkpoint. The construction up to this point is local and exact-entry dependent only:

$$(H, p) \longmapsto L_{\text{BR}}^\dagger(D, N, p, H).$$

It does not use terminal-ladder depth, the LOW continuation tree, or $B_{\text{hard}}^\dagger$. Those objects are constructed only below, after $L_{\text{BR}}^\dagger$ is available.

16 Terminal high tails and the last-occurrence ladder

Consider the suffix after the last LOW cut. If it is nonempty, the deterministic run enters height

$$b = B_0 + 1$$

and never again visits height at most B_0 . Let its final height be $h_f \geq b$. For every $\ell \in \{b, \dots, h_f\}$ let c_ℓ be the last cut in the terminal tail at deterministic height ℓ .

Lemma 16.1 (Shape of one ladder step). *For $\ell < h_f$, the factor from c_ℓ to $c_{\ell+1}$ consists of one +1 deterministic step followed by a same-level bounded-return excursion at base $\ell + 1$.*

Proof. Immediately after c_ℓ the deterministic effect cannot be 0, because that would create a later occurrence of level ℓ , and cannot be -1 , because reaching the final height would force another occurrence of level ℓ . Thus it is $+1$. Before $c_{\ell+1}$ the run cannot go below $\ell + 1$ and ends there by definition of the last occurrence. $\square$

At c_ℓ record

$$X_\ell = (p_\ell, H_\ell),$$

where H_ℓ is the full exact N forward envelope. From a concrete X_ℓ , enumerate the finitely many $+1$ letters and then all same-level return words allowed by Theorem 15.6. Thus the ladder successor relation is effectively finite.

Theorem 16.2 (Ladder deletion). *If $\ell < \ell'$ and*

$$p_\ell = p_{\ell'}, \quad H_\ell \leq H_{\ell'}$$

componentwise, deleting the whole factor from c_ℓ to $c_{\ell'}$ yields a shorter word still in $L(D) \setminus L(N)$.

Proof. The deleted factor has deterministic net effect $e = \ell' - \ell > 0$. By last-occurrence minimality, after $c_{\ell'}$ the original deterministic suffix never goes below ℓ' . After deletion the same suffix is shifted down by e and therefore never goes below $\ell \geq 0$; the equal control state rejoins the accepting suffix. On the N side, the original rejection at $c_{\ell'}$ gives $H_{\ell'} < R_{\ell'}$, hence $H_{\ell} \leq H_{\ell'} < R_{\ell'}$, so the shortened word remains rejected. $\square$

The ladder state space is a finite disjoint union of copies of $\mathbb{N}^{|Q_N|}$. A shortest counterexample therefore yields a bad sequence. Together with the effective finite successor relation, Dickson plus König gives a finite effectively constructible bad-sequence tree.

Theorem 16.3 (Terminal compactness). *For every concrete high exit (p, H) there is a total computable*

$$L_{\text{term}}^{\dagger}(D, N, p, H)$$

that bounds the terminal high tail of a shortest hard-direction counterexample, except for bounded decisive continuations that are charged separately.

Proof. From one exact ladder state, the finite alphabet and Theorem 15.6 give an effectively finite successor set: one feasible +1 spine letter followed by one bounded same-level return factor. By Theorem 16.2, a shortest terminal tail contains no comparable pair of ladder states. Hence every branch of the generated ladder tree is a Dickson bad sequence. The tree is effectively finitely branching, so König implies that it is finite; breadth-first generation therefore terminates. Summing the literal edge lengths along this finite tree and taking the maximum gives $L_{\text{term}}^{\dagger}$. This construction calls $L_{BR}^{\dagger}$ but neither the LOW tree nor the final hard bound. $\square$

17 The exact LOW continuation tree and the hard small-witness theorem

At every LOW cut record the exact state

$$X = (p, d, H) \in Q_D \times \{0, \dots, B_0\} \times (\mathbb{N} \cup \{\perp\})^{Q_N}.$$

Encode $\perp$ by 0 and a finite value h by $h + 1$. Order LOW states by

$$(p, d, H) \preceq (p', d', H') \iff p = p', d = d', H \leq H'$$

componentwise.

Lemma 17.1 (Exact LOW deletion). *A shortest $D \setminus N$ witness contains no two LOW cuts $i < j$ with $X_i \preceq X_j$.*

Proof. Equal p, d means that the deterministic physical configurations coincide, so deleting the interval rejoins the identical accepting suffix. If R_j is the exact backward requirement of that suffix, rejection gives $H_j < R_j$; since $H_i \leq H_j$, the shortened word remains rejected. $\square$

17.1 Node-dependent successor enumeration

For one exact LOW state $X = (p, d, H)$ define $\text{Succ}^{\dagger}(X)$ as follows. If $d < B_0$, or $d = B_0$ and the next deterministic effect is nonpositive, enumerate all feasible one-letter LOW successors. If $d = B_0$ and a feasible letter a has effect +1, compute the post-exit control and exact envelope

$$(p^+, H^+) = (p^+, T_a(H)).$$

If the actual run later returns to LOW for the first time, the high interior before the final -1 return step is a same-level bounded-return factor. Enumerate every D-feasible candidate interior of length at most

$$L_{BR}^{\dagger}(D, N, p^+, H^+)$$

and every feasible final -1 letter, and compute the resulting exact LOW envelope. If no LOW return occurs, the branch is terminal and belongs to Section 16. If a bounded decisive continuation is produced, record its remaining length from the current node.

Theorem 17.2 (Effective LOW successors). *For every exact LOW state X , $\text{Succ}^\dagger(X)$ is finite and effectively enumerable, and contains the next LOW state of every shortest hard-direction counterexample whenever such a next LOW cut exists.*

Proof. The one-letter LOW cases are finite. In the $+1$ exit case the concrete exact post-exit envelope $H^+ = T_a(H)$ is computable, and Theorem 15.6 supplies a finite length bound for every retained same-level interior before the first return. Exhaustive enumeration up to that bound, followed by one feasible -1 return letter, therefore lists every possible next LOW state. Shortening and promise-contradiction outcomes cannot occur on the retained shortest promised branch; bounded decisive completions are recorded separately. No quantity defined from the later LOW tree is used in this enumeration. $\square$

17.2 Finite bad-sequence tree

Build $T_{\text{low}}^\dagger$ from the exact initial LOW state. A node is a finite sequence

$$X_0, \dots, X_k$$

with $X_{i+1} \in \text{Succ}^\dagger(X_i)$ and no comparable pair $X_i \preceq X_j$ for $i < j$. Attach to every edge its literal realization length and let $w(X)$ be the accumulated path weight to node X .

Theorem 17.3 (Finite exact-LOW continuation tree). *The tree $T_{\text{low}}^\dagger(D, N)$ is finite and effectively constructible.*

Proof. Every node is effectively finitely branching by Theorem 17.2. Every branch is a bad sequence in a finite disjoint union of copies of $\mathbb{N}^{Q_N}$, so Dickson's lemma excludes infinite branches. König's lemma therefore makes the finitely branching tree finite. Breadth-first exhaustive generation terminates because every successor set is explicitly computable. $\square$

17.3 Prefix-safe decisive accounting

For a tree node X , let $C_{\text{dec}}(X)$ be the maximum remaining continuation length from X among genuinely bounded decisive completions produced by the local analysis; set it to 0 if none occur. Define

$$B_{\text{dec}}^\dagger(D, N) = \max_{X \in T_{\text{low}}^\dagger} (w(X) + C_{\text{dec}}(X)),$$

with the obvious $+1$ correction when a decisive continuation begins after a high-exit letter. This explicit prefix term is essential: a bounded completion from a concrete node is not automatically a bounded complete word from the initial configuration.

Define

$$B_{\text{low}}^\dagger = \max_{X \in T_{\text{low}}^\dagger} w(X).$$

For every node $X = (p, B_0, H)$ and every feasible $+1$ exit letter a , compute the terminal bound from Section 16 and set

$$B_{\text{term}}^\dagger = \max \left\{ w(X) + 1 + L_{\text{term}}^\dagger(D, N, p^+, T_a(H)) \right\}.$$

Finally put

$$B_{\text{tree}}^\dagger = \max \{ B_{\text{low}}^\dagger, B_{\text{term}}^\dagger, B_{\text{dec}}^\dagger \}.$$

As an independent bounded-root cross-check retain

$$B_{\text{nr}}(D, N) = F(H_* - 1, D, N), \quad H_* = B_F^2 + 2B_F + 1,$$

and define

$$B_{\text{hard}}^\dagger(D, N) = \max\{B_{\text{tree}}^\dagger(D, N), B_{\text{nr}}(D, N)\}.$$

Theorem 17.4 (Hard-direction small-witness theorem). *Under the promise*

$$L(N) \subseteq L(D),$$

if

$$L(D) \not\subseteq L(N),$$

then there exists

$$w \in L(D) \setminus L(N)$$

with

$$|w| \leq B_{\text{hard}}^\dagger(D, N).$$

Moreover $B_{\text{hard}}^\dagger$ is a total computable function of finite encodings of D, N .

Proof. Choose a shortest counterexample. At a current LOW cut, either the word ends; the next letter remains LOW; the next letter exits high and later returns to LOW; or it exits high and never returns. These cases are exhaustive. The first is charged to $B_{\text{low}}^\dagger$; the second and third follow edges of $T_{\text{low}}^\dagger$ by Theorem 17.2; and the fourth is charged to $B_{\text{term}}^\dagger$. Any endpoint-preserving shortening is impossible by shortestness, any promise-forbidden branch is impossible by $L(N) \subseteq L(D)$, and every genuinely bounded decisive continuation is charged to $B_{\text{dec}}^\dagger$ including the prefix weight. Thus the shortest counterexample is bounded by $B_{\text{tree}}^\dagger$, and hence by $B_{\text{hard}}^\dagger$. $\square$

Corollary 17.5 (Hard inclusion under the promise). *Under $L(N) \subseteq L(D)$, the inclusion $L(D) \subseteq L(N)$ is decidable.*

Proof. Compute $B_{\text{hard}}^\dagger$, enumerate all words up to this length, and test membership in D and N . If none lies in $L(D) \setminus L(N)$, Theorem 17.4 implies the inclusion. $\square$

Final local-to-global dependency audit. The only dependency from Sections 16–17 back into the bounded-return analysis is the already computed local function $L_{BR}^\dagger$. The terminal ladder first constructs $L_{\text{term}}^\dagger$ from finitely many +1 letters and $L_{BR}^\dagger$ -bounded same-level returns. Exact LOW successors then use only the concrete envelope $H^+ = T_a(H)$ together with these two local functions. Dickson plus König constructs $T_{\text{low}}^\dagger$ only after every node has its finite successor set. Finally $B_{\text{low}}^\dagger$, $B_{\text{term}}^\dagger$, $B_{\text{dec}}^\dagger$ and $B_{\text{hard}}^\dagger$ are maxima over the finished finite tree. Thus the dependency chain is

$$L_{BR}^\dagger \longrightarrow L_{\text{term}}^\dagger \longrightarrow \text{Succ}^\dagger \longrightarrow T_{\text{low}}^\dagger \longrightarrow B_{\text{hard}}^\dagger,$$

and no later global bound occurs in the construction of an earlier object.

18 The known deterministic-right inclusion

Our acceptance convention is finite-word coverability/final-state semantics: the counter must remain nonnegative at every prefix and the run must end in a designated final control state; the final counter value need not be zero. Hofman and Totzke prove deterministic-right OCN trace inclusion decidable and explain that their compact-witness method applies to monotone relations including languages defined by final-state acceptance [1]. Totzke later stated explicitly that the upper-bound argument works analogously for coverability semantics [5]. Only decidability is used here.

Theorem 18.1 (Deterministic-right final-state inclusion [1]). *For a nondeterministic OCN N and a deterministic OCN D , it is decidable whether*

$$L(N) \subseteq L(D)$$

under the finite-word final-state/coverability semantics used in this paper.

Remark 18.2. The orientation is essential: the external theorem supplies $N \subseteq D$, not the opposite standalone inclusion. Our initial counter value 0 is a special case of the fixed/encoded initial values allowed in the deterministic-right framework. For an arbitrary integer-weight input, Lemma 2.3 is applied first to both nets; it preserves the alphabet, final-state language, and determinism status, after which this unit-step theorem applies. No complexity bound for the complete equivalence procedure is inferred from the trace-inclusion complexity result.

19 Equality

Proof of Theorem 1.1. Apply Lemma 2.3 to the two input nets, obtaining a deterministic unit-step net D^u and a nondeterministic unit-step net N^u with

$$L(D^u) = L(D), \quad L(N^u) = L(N).$$

It therefore suffices to decide equality for the normalized pair. First decide

$$L(N^u) \subseteq L(D^u)$$

using Theorem 18.1. If this fails, the languages are unequal. Otherwise the reverse-inclusion promise holds for the normalized pair, and Corollary 17.5 decides

$$L(D^u) \subseteq L(N^u).$$

By language preservation of the normalization, the original languages are equal exactly when both normalized inclusions hold. $\square$

20 Related work and provenance

Deterministic-right inclusion for one-counter nets is due to Hofman and Totzke [1]; their result also provides the external inclusion orientation used in Section 18. Deterministic one-counter automata admit efficient equivalence checking [2], while history-deterministic one-counter nets form a broader comparison class with additional decidability properties [3].

The mixed deterministic/nondeterministic OCN language-equivalence question appears as Problem 24.4(B) in the Automata Exchange collection [4]. The problem record accessed in September 2026 still lists this question as open, and Totzke’s 2023 discussion describes the same status while distinguishing trace, coverability, and reachability semantics [5]. Accordingly, the theorem proved here should be regarded as a solution candidate for that listed problem pending

independent expert verification and the usual publication process; no claim of external validation or priority is made by the present paper.

The research lineage from the earlier 2:1/p:q normalization program through the Prime-DOCA and Two-Factor quotient/decomposition work is summarized in the Introduction [6, 7, 8, 11, 9, 10]. Those works motivate the residual and canonical-cut viewpoint but are not logical dependencies of the theorem proved here.

21 Discussion

The proof is effective but intentionally complexity-agnostic. The hard-direction bound is obtained by nested finite constructions and bad-sequence trees; optimizing it is a separate problem.

The decisive structural point is that no single selected nondeterministic run is allowed to stand in for the full envelope. Triple-product differential potentials and the reduced useful-mode graph control arbitrary mode switching first. Only then do exact active-color recurrence and deterministic same-level balance bound the canonical event trace; source–sink accounting remains local to the route reductions where it is actually needed. At the global level, absolute counter growth is not forced into a finite-state quotient: unmatched growth is carried by the terminal level ladder, while repeated visits to the bounded band are controlled by the exact forward-envelope WQO.

The natural next problem is the corresponding comparison in which the deterministic machine has zero tests. Positive-counter intervals still inherit much of the theory above, but zero-test visits destroy the global monotonicity used by the present LOW/ladder deletion arguments.

A Effectivity ledger

Every numerical object used in the final algorithm is generated by a terminating finite or well-quasi-order construction.

- (E1) $F(B, D, N)$ is obtained by exhaustive generation of the finitely branching bounded-height bad-sequence tree.
- (E2) Primitive pair/triple cycle universes, connector colors, boundary signatures, anchored colors, block graphs, critical potentials, rational slope data, and cone-separation witnesses are finite and effectively enumerable.
- (E3) For every deterministic entry bound B , the quantitative threshold $R_{\text{rel}}(D, N, B)$ of Corollary 12.6 is computable as a finite maximum of synchronized SCC cone-separation constants, condensation-DAG depths, and bounded connector/simple-path errors.
- (E4) B_{src} and the bounded local episode constants are maxima over finitely many bounded core/SCC types.
- (E5) The triple-product one-sign potentials and mixed-sign bounded packages are computed on the finite graph $Q_D \times Q_N \times Q_N$.
- (E6) For each concrete bounded-return entry, the finite raw graph M_{raw} is reduced by the explicit (R1)–(R5) operation to M_{red} . Only M_{red} carries the no-positive-cycle theorem. Its N-potential is computed SCC-by-SCC, with a finite condensation-DAG correction and exact bounded scalar labels on retained deferral edges.
- (E7) Neutral power caps $J_x(D, N, H)$ are found by enumerating the exact forward orbit until a Dickson comparable pair occurs.

- (E8) Negative-mean kill thresholds use computable maximum-cycle-mean gaps. In zero-mean regions, critical relation monoids are used only to identify periodic relation states: zero-D repetition is controlled by exact-entry WQO/deletion, while positive-D zero-mean repetition is retained as deterministic funding and transferred to later chronological exposure. Exact-slope source–sink shortening uses balanced critical periods together with bounded adjusted budgets.
- (E9) $L_{BR}^\dagger(D, N, p, H)$ is computed in the acyclic order $C_{BR} \rightarrow B_H \rightarrow (G_{col}, L_{edge}, D_{edge}) \rightarrow K_{evt} \rightarrow L_{BR}^\dagger$. The event budget K_{evt} follows from deterministic cycle erasure: every retained exact-color cycle has positive integral D effect, while the same-level route has net D effect zero and its residual simple color path has bounded negative D effect. Thus no cumulative N-loss assumption is used.
- (E10) Terminal ladder successors are finitely enumerated using $L_{BR}^\dagger$, and the finite bad-sequence ladder tree yields $L_{term}^\dagger$.
- (E11) LOW successors are finitely enumerated node by node. Exhaustive generation of $T_{low}^\dagger$ terminates by Dickson plus König. Only after the tree is complete are the weighted maxima $B_{low}^\dagger$, $B_{term}^\dagger$, and $B_{dec}^\dagger$ taken.
- (E12) The final hard bound is explicitly

$$B_{hard}^\dagger = \max\{B_{tree}^\dagger, F(H_* - 1, D, N)\}.$$

No uniform global high-excursion bound, uniform terminal-envelope bound, or bounded inherited suffix is assumed in its own construction.

B Why absolute backward requirements are not finite-state data

Absolute backward requirements may be arbitrarily large. The proof never stores them globally. In low-active regions, the exact finite color

$$\chi(t) = (p_D(t), A_t, H_t|_{A_t})$$

is sufficient to force deletion or positive-return recurrence without bounding the values of the requirement vector. In zero-mean critical regions, only a finite large-deficit/bounded-deficit relation state is retained. In source–sink localization, escaping endpoint credit is converted into a bounded-entry relative-source episode. The global WQO state stores the exact *forward* envelope H , not the absolute backward vector R .

C A note on overlapping events

Primitive bounded event words may overlap. Fix a canonical extraction rule: choose the earliest-starting exposed event and break ties by a fixed order on the finite event universe. Overlap-connected fragments have bounded local word length; if overlaps propagate through a longer region, their successive canonical endpoints themselves form the bounded-step event sequence used in the full-envelope excursion analysis. Exposure-free gaps therefore still obey the connector bound.

Statements and Declarations

Author contributions. Alp Eren Bütün is the sole author and was responsible for conceptualization, methodology, formal analysis, investigation, theorem development, computational validation where reported, visualization, writing—original draft, and writing—review and editing.

Funding. No funding was received for conducting this study.

Competing interests. The author has no relevant financial or non-financial interests to disclose.

Data and code availability. The paper is theoretical and uses no empirical dataset. All new definitions, constructions, formulas, finite certificates, and proofs introduced for the hard inclusion are contained in the manuscript. The sole external decision-theoretic input to the final equivalence algorithm is Theorem 18.1, the deterministic-right inclusion result cited there. Any computations reported in the manuscript are independent regression checks and are not proof premises. No external dataset or executable code is required to establish the new arguments.

Ethics approval and consent to participate. Not applicable.

Consent for publication. Not applicable.

References

- [1] P. Hofman and P. Totzke, Trace inclusion for one-counter nets revisited, *Theoretical Computer Science* 735 (2018), 50–63. doi:10.1016/j.tcs.2017.05.009.
- [2] S. Böhm, S. Göller, and P. Jančar, Equivalence of deterministic one-counter automata is NL-complete, in *Proceedings of the 45th Annual ACM Symposium on Theory of Computing (STOC 2013)*, pp. 131–140, ACM, 2013. doi:10.1145/2488608.2488626.
- [3] K. Prakash and K. S. Thejaswini, On history-deterministic one-counter nets, in *Foundations of Software Science and Computation Structures (FoSSaCS 2023)*, Lecture Notes in Computer Science, vol. 13992, pp. 218–239, Springer, 2023. doi:10.1007/978-3-031-30829-1_11.
- [4] Automata Exchange, Problem 24.04: Language equivalence between nondeterministic and deterministic one-counter machines, <https://github.com/michaelcadilhac/automata-exchange/blob/main/24.04-language-equivalence-between-nondeterministic-and-deterministic-one-counter-machines.org>, accessed September 16, 2026.
- [5] P. Totzke, Answer to “Language equivalence between deterministic and non-deterministic counter net,” MathOverflow, 2023, <https://mathoverflow.net/questions/451272/language-equivalence-between-deterministic-and-non-deterministic-counter-net>, accessed September 16, 2026.
- [6] A. E. Bütün, Completion Frontiers and Quotient Geometry in Deterministic Pushdown Automata for Rational Binary Counting, Zenodo, 2026. doi:10.5281/zenodo.22709361.
- [7] A. E. Bütün, Real-Time p;q Deterministic Pushdown Automata: Generalization, Normalization, and Exact Control Width — A WRITE-SEARCH-CANCEL Theory for Weighted Binary Balance Languages, Zenodo, 2026. doi:10.5281/zenodo.22709331.
- [8] A. E. Bütün, Semantic Uniqueness and LIFO Lift Theory for Real-Time Pushdown Normalization, Zenodo, 2026. doi:10.5281/zenodo.22709334.
- [9] A. E. Bütün, State-Primality Is Undecidable for Deterministic One-Counter Automata, Zenodo, 2026. doi:10.5281/zenodo.22709326.
- [10] A. E. Bütün, Two-Factor Decomposition of Deterministic One-Counter Automata Is Undecidable, Zenodo, 2026. doi:10.5281/zenodo.22709338.
- [11] A. E. Bütün, Deterministic Pushdown Normalization, Zenodo, 2026. doi:10.5281/zenodo.22712151.